

M Pharmacy Notes

Cosmetics and Cosmeceuticals

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Syllabus: -

Unit I

Cosmetics - Regulatory: Definition of cosmetic products as per Indian regulation. Indian regulatory requirements for labeling of cosmetics. Regulatory provisions relating to import of cosmetics. Misbranded and spurious cosmetics. Regulatory provisions relating to manufacture of cosmetics - Conditions for obtaining license, prohibition of manufacture and sale of certain cosmetics, loan license, offences and penalties.

Unit II

Cosmetics - Biological aspects: Structure of skin relating to problems like dry skin, acne, pigmentation, prickly heat, wrinkles and body odor. Structure of hair and hair growth cycle. Common problems associated with oral cavity. Cleansing and care needs for face, eye lids, lips, hands, feet, nail, scalp, neck, body and under-arm.

Unit III

Formulation Building blocks: Building blocks for different product formulations of cosmetics/cosmeceuticals. Surfactants - Classification and application. Emollients, rheological additives: classification and application. Antimicrobial used as preservatives, their merits and demerits. Factors affecting microbial preservative efficacy. Building blocks for formulation of a moisturizing cream, vanishing cream, cold cream, shampoo and toothpaste. Soaps and syndet bars.

Perfumes; Classification of perfumes. Perfume ingredients listed as allergens in EU regulation.

Controversial ingredients: Parabens, formaldehyde liberators, dioxane.

Unit IV

Design of cosmeceutical products: Sun protection, sunscreens classification and regulatory aspects. Addressing dry skin, acne, sun-protection, pigmentation, prickly heat, wrinkles, body odor., dandruff, dental cavities, bleeding gums, mouth odor and sensitive teeth through cosmeceutical formulations.

Unit V

Herbal Cosmetics: Herbal ingredients used in Hair care, skin care and oral care. Review of guidelines for herbal cosmetics by private bodies like cosmos with respect to preservatives, emollients, foaming agents, emulsifiers and rheology modifiers. Challenges in formulating herbal cosmetics.

Unit I - Cosmetics - Regulatory

Cosmetics are regulated under The Drug and Cosmetic Act (D & C Act) 1940 and The Rule in 1945 and the labeling Declarations by Bureau of Indian Standards (BIS). BIS sets the standards for cosmetics for the products listed under Schedule 'S' of the Drugs and cosmetics Rules 1945s.

According to D & C Act Cosmetic is define as the means any article intended to be rubbed, poured, sprinkled or sprayed on, or introduced into, or otherwise applied to, the human body or any part thereof for cleansing, beautifying, promoting attractiveness, or altering the appearance, and includes any article intended for use as a component of cosmetic.

Chapter III of Drug and Cosmetic Act deals with import of Drug and Cosmetic.

Now import of certain cosmetic is prohibited under section 10 of the Act and rules 134A, 135, 135A of Drug and Cosmetic Rules 1945 they are:

- Any cosmetic which is not of standard quality,
- Any misbranded or spurious cosmetic,
- Any cosmetic containing any ingredient which may render it unsafe or harmful for use under the directions indicated or recommended,
- Any cosmetic the import of which is prohibited by rule,
- Any cosmetic containing hexachlorophene,
- Any cosmetic in which lead and arsenic compound has been used for coloring purpose,
- Any cosmetic which contains mercury compound.

Indian regulatory requirements for labeling of cosmetics

Rule of The Act C & D Divided in Different Parts:

- Part 13 - import of cosmetics.
- Part 14 - manufacturing of cosmetics.
- Part 15 - labeling, packaging, and standard of cosmetics.
- Part 15-A - approval of institute for analysis of drug, cosmetics & raw material.

Labeling declaration

- a. Name and address of the importer;
- b. Generic or common name of the commodity packed;
- c. Net quantity in terms of standard unit of weights and measures;
- d. Month and year of packing in which the commodity is manufactured or packed or imported; and
- e. Maximum retail sales price (MRP) at which the commodity in packaged form may be sold to the end consumer.

On both the inner & outer labels:-

- a. Name of the cosmetic.
- b. Name of the manufacturer and complete address of the premises of the manufacturer where the cosmetic has been manufactured.

On the outer label

- a. A declaration of the net contents expressed in terms of weight for solids/ semi solids, fluid measure for liquids. (Note: In the case of package of perfume, toilet water or the like, the net content which does not exceed 60 ml or any package of solid or semi solid cosmetic

where the net content doesn't exceed 30 gms, the statement of net content need not appear).

- b. The names of ingredients in the order of percentage of content.

On the inner label

- a. Where a hazard exists, adequate directions for safe use, warning, caution or special direction required to be observed by the consumer, statement of the names and quantities of the ingredients that are hazardous or poisonous.
- b. The names of ingredients in the order of percentage of content.
- Batch number preceded by the letter "B"
 - Manufacturing license number preceded by the letter "M."

Regulatory provision relating to import of cosmetics:-

Import of certain cosmetics is prohibited under D & C Act & Rules 134A, 135, 135A.

- Any cosmetic this is not of std quality.
- Any cosmetic that is misbranded or spurious.
- A cosmetic containing ingredient which render it injurious or harmful or unsafe to use.
- A cosmetic whose import is prohibited by rule.
- Any cosmetic except toilet soap containing hexachlorophene.
- Any cosmetic in which lead or arsenic compound used for coloring purpose.
- Any cosmetic which contain mercury compounds.

In eighties, stds formed by Indian std institution were adopted for certain cosmetics. 28 cosmetics have been placed under Schedule 'S' to the rules & are required to comply with Indian stds. & these are:

- Skin powder
- Tooth powder
- Shampoos, soap based
- Skin creams
- Hair creams
- Cologne
- Aftershave lotions
- Depilatories, chemical
- Cosmetic pencils
- Toilet soap
- Baby toilet soap
- Shaving soap

- Skin powder for infants
- Powder hair dye
- Tooth paste
- Hair oils
- Kumkum powder
- Shampoos, synthetic detergent based
- Oxidation hair dyes, liquid
- Nail polish
- Pomades & brilliantine
- Shaving creams
- Lipsticks
- Henna powder
- Liquid toilet soaps
- Transparent toilet saps

Misbranded & spurious cosmetics have been defined in section 17-C & 17-D as under:

Misbranded cosmetics: Any cosmetic

- a. If it contain a color which is not prescribed
- b. If it not labeled in the prescribed manner
- c. If the label or container bears any statement which is false or misleading in any particular.

Spurious cosmetic: Any cosmetic

- a. If it imported under a name which belongs to another cosmetic.
- b. If the label or container bears the name of an individual or a company purporting to be the manufacturer of the cosmetic which individual or company is fictitious or does not exist,
- c. If it purports to be the product of a manufacturer of whom it is not truly a product.

Misbranded Cosmetic (section 17-C and 17-D)

A cosmetic is deemed to be misbranded if:

- a. If it contains a color which is not prescribed; or
- b. If it is not labeled in the prescribed manner; or
- c. If the label or container bears any statement which is false or misleading in any particular.

Spurious Cosmetic:

A cosmetic is deemed to be spurious if:

- a. If it is imported under a name which belong to another cosmetic; or
- b. If the label or containers bears the made of an individual or a company purporting to be the manufacturer of the cosmetic which individual or company is fictitious or does not exist; or
- c. If it purports to be the product of a manufacturer of whom it is not truly a product.

Restriction

Rule 134A also prescribes that no cosmetic shall be imported which contains a coal tar color other than one prescribed in schedule Q and Indian standards (IS: 4707 Part I) to the above rules and coal tar color used in cosmetic should not contain more than -

- a. 2 ppm of arsenic as arsenic trioxide.
- b. 20 ppm of Lead.
- c. 100 ppm of heavy metals other than lead calculated as total of the respective metals.

Regulatory provision relating to manufacture of cosmetics

How to obtain license

The procedure should be followed in order to manufacture cosmetics in India has been laid down under the D & C Rules, 1945.

Schedule M-II classifies cosmetics into 11 broad product categories:

1. Powders
 2. Creams, lotions, emulsions, pastes, cleansing milks, shampoos, pomade, brilliantine, shaving creams, hair oils, etc.
 3. Nail polishes & nail lacquers
 4. Lipsticks & lip gloss
 5. Depilatories
 6. Preparations used for eyes
 7. Aerosol
 8. Alcoholic fragrance solutions
 9. Hair dyes
 10. Tooth powders & tooth pastes
 11. Toilet soaps.
- The application has to be submitted in form (form-31) along with a license fee of Rs. 2500 & an inspection fee of Rs. 1000.
 - The manufacturer has to ensure that the production is done in presence of a competent & qualified technical staff & at least one of the staff persons should possess the following educational requirements:
 1. Holds a Diploma in Pharmacy approved by the PCI under the Pharmacy Act 1948; or
 2. Is registered under the Pharmacy Act, 1948; or

3. Has passed the intermediate examination with chemistry as one of the subjects or any other examination as recognized by the licensing authority as equivalent to it.

Before granting or refusing the license, the licensing Authority is required to order inspection of the whole premises where the operations are to be carried out. The inspectors are appointed under the act, they are then required to submit a detailed report to the Licensing Authority which can then decide whether to grant the license or not.

Schedule M (II) of D & C rule 1945 deals with requirements for the factory premises for the manufacture of cosmetics:

General requirements:

1. Location & surroundings
2. Buildings
3. Water supply
4. Disposal of water
5. Health clothing & sanitary requirements of the staff
6. Medical devices
7. Working benches for the operation of cosmetics.

Requirement of plant & equipment: Depending on the manufacture of different formulation equipment, area & other requirements are been taken into consideration.

A. Powder:

- a. Powder mixer of suitable type provided with a dust collector
- b. Perfume & color blender
- c. Ball mill or suitable grinder
- d. Filling & sealing equipment

- e. Weighing & measuring devices
- f. Storage tanks
- g. An area of 15 square meters is recommended.

B. Creams, Lotions, Shampoos, Hair Oils:

- a. Mixing & storage tanks of suitable materials.
- b. Suitable agitator
- c. Heating kettle steam gas or electrically heated.
- d. Filling & sealing equipment
- e. Weighing & measuring devices
- f. An area of 25 square meters is recommended.

C. Nail Polishes & Nail Lacquers:

- a. **Equipment:** A suitable mixer, storage tanks, filling & sealing machine, weighing & measuring devices.
- b. **Premises:** should be suitable in an industrial area, be separate from other cosmetic shall manufacturing areas, floors, walls & doors shall be proof,.
- c. **Storage:** all explosive solvent & ingredients shall be stored in metal cupboards.
- d. **Manufacture:** workers shall be asked to wear shoes in the section.

D. Lipsticks:

- 1. Vertical mixer
- 2. Mixing vessel
- 3. Triple roller mill
- 4. Weighing & measuring devices
- 5. An area of 15 square meters is recommended.

E. Hair Dyes:

- a. Stainless steel tanks
- b. Mixer
- c. Filling & sealing unit
- d. Weighing & measuring devices
- e. Gloves, masks.
- f. An area of 15 square meters is recommended.

F. Tooth Powder:

- a. Weighing & measuring devices
- b. Dry mixer
- c. Stainless steel sieves
- d. Powder filling & sealing unit
- e. Area of 15 square meters is recommended.

G. Tooth Pastes:

- a. Weighing & measuring devices
- b. Kettle steam, gas or electrically heated
- c. Planetary mixers
- d. Tube filling equipment
- e. Crimping machine
- f. An area of 15 square meters is recommended.

H. Aerosol:

- a. Air compressor
- b. Mixing tanks
- c. Filtering, filling & cramping

Equipment

- a. Liquid filling unit
- b. Leak test equipment

- c. Fire extinguisher
- d. An area of 15 square meters is recommended.

I. Eye Brows, Eye Lashes, Eye Liners:

- a. Mixing tanks
- b. Filtering, filling & sealing equipment.
- c. Weighing & measuring devices
- d. An area of 15 square meters is recommended.

J. Alcoholic Fragrance Solutions:

- a. Mixing tanks
- b. Filtering, filling & sealing equipment
- c. Weighing & measuring equipment
- d. An area of 15 square meters

Conditions of license:

- Manufacturer shall have to continue to provide & maintain the premises, equipment & staff as at the time of grant of license.
- Manufacturer should test each batch of raw material before its use in the manufacture of cosmetics.
- Manufacturer should have to test each batch of finished cosmetics.
- Should have to maintain records of raw materials, manufacturer of each batch of cosmetic & their testing.

Particulars to be shown in the manufacturing records:

1. Serial no.
2. Name of the product
3. Batch size
4. Batch no.
5. Name of all ingredients used

6. Actual production & packaging particulars indicating the size & qt of finished packing.
7. Signature of the expert staff responsible for the manufacture.

The requirements pertaining to labeling of products:

Following are some of the labeling requirements as laid down under the D & C Rules, 1945:

1. Name of the product along with the manufacturing address must be mentioned on both the inner & the outer labels. If the container is small in size, then the principal place of mfg & the pin code are enough.
2. The outer label should clearly specify the net contents of the ingredients used in the manufacture of the product.
3. The inner label should contain the directions for use along with any warning or caution that may be necessary. It should also contain names & quantities of ingredients which are hazardous in nature.
4. A distinctive batch no. preceded by letter 'B' along with mfg license no. preceded by letter 'M' must be mentioned on the label.
5. Quality standards in case of the aforementioned categories of products must conform to the Indian standards laid down & revised by the BIS.

Prohibition of Manufacture

- Drugs not of the standard quality or misbranded, adulterated or spurious.
- Patent or proprietary medicine.
- Drugs in schedule J.
- Risky to human beings or animals.
- Drugs without therapeutic value.
- Preparation containing cyclamates.

Prohibition of following classes of the drugs:

- Misbranded or spurious cosmetics and of substandard quality
- Cosmetics containing hexachlorophene or mercury compounds
- Cosmetics containing color which contain more than -
 - 2ppm of arsenic
 - 20 ppm of lead
 - 100 ppm of heavy metals
- Eye preparations containing coal-tar color

Class of Drug Prohibited to Sale

- Misbranded, spurious, adulterated, drugs not of standard quality
- Patent / propriety drugs with undisclosed formula.
- Schedule J drugs.
- Expired drugs.
- Drugs used for consumption by government schemes such as, Armed forces.
- Physician's samples.

Loan License

Definition

A person (applicant) who does not have his own arrangements (factory) for the manufacturing but who wish to manufacturing facilities owned by another licensee. Such licenses are called as Loan licenses.

Loan license is issued for;

1. Drugs other than specified in Schedule C/C1 & X.
2. Drugs specified in schedule C/C1.

Offences and Penalties

OFFENCES	PENALTIES
Manufacturing of any spurious drugs	A) 1-3 Years imprisonment & Rs. 5000 fine B) B) 2-6 years imprisonment & Rs. 10000 fine on the subsequent conviction
Manufacturing of adulterated drugs	A) 1 year imprisonment & Rs. 2000 fine B) B) 2 years imprisonment & Rs.2000 fine on the subsequent conviction
Manufacturing of drugs in contravention of the provisions	A) Imprisonment up to 3 months & Rs 500 fine B) B) Imprisonment up to 6 months & Rs.1000 fine .

Unit 2 - Cosmetics - Biological Aspects

Introduction

Skin is the largest body organ which weighs about 4kg and an area of 2m².

It performs many vital functions, including protection against external physical, chemical, as well as prevention of excess water loss from the body and a role in thermoregulation.

Three layers of skin:

1. Epidermis
2. Dermis
3. Hypodermis

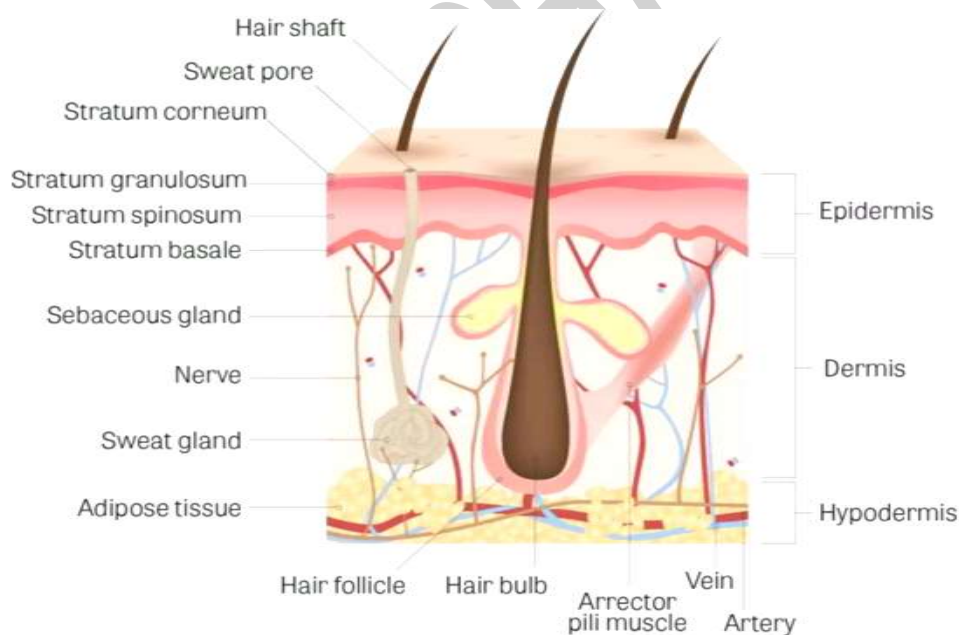


Fig: Microscopic cross-section of skin.

1. Epidermis

- Epidermis is primarily made up of stratified squamous epithelium.
- Epidermis doesn't contain any blood vessels.
- Epidermis thickness ranges from less than 0.1mm (on eyelids) to 1 mm (on palm and sole), the thickness is kept constant by cells dividing in the deepest layer (basal or germinative layer).

Layers of Epidermis - Epidermis divided into different regions or strata:

- Stratum corneum
- Stratum lucidum
- Stratum granulosum
- Stratum spinosum
- Stratum basale (Stratum germinativum)

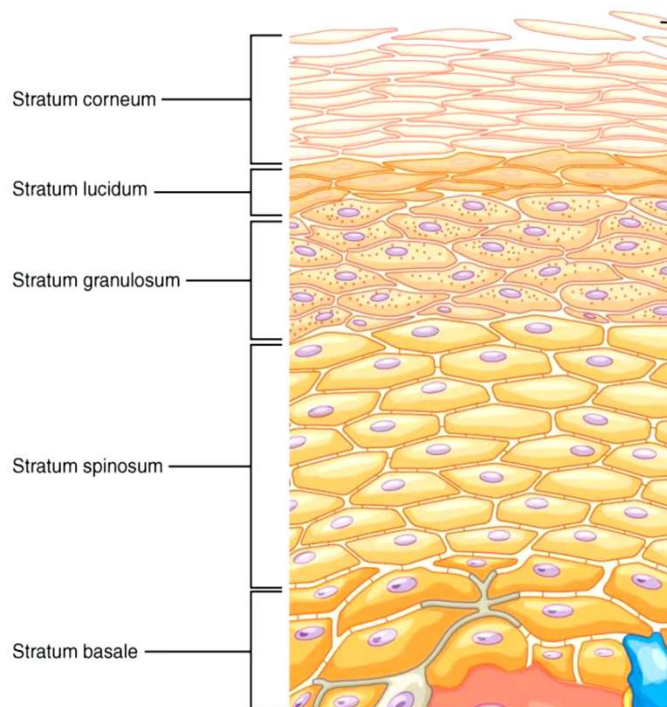


Fig: Microscopic layers of epidermis.

a. Stratum Corneum

- Its most superficial layer of skin consisting of dead cells corneocytes.
- The stratum corneum is the rate limiting barrier that restricts the inward & outward movement of chemical substances.
- This layer is responsible for the "spring back" or stretchy properties of skin.
- The stratum corneum exhibit regional difference in thickness over the body.
- It is thick on the palm of the hand & soles of the feet in an adult.
- Stratum corneum contains a dense network of keratin, a protein that helps keep the skin hydrated by preventing water evaporation.
- These cells can also absorb water, further aiding in hydration.

b. Stratum lucidum

- Stratum lucidum is seen in thick skin of the palms and soles of feet.
- It contains 3-5 rows of clear flat dead cells.
- Its mainly composed of 'keratohyalin' which is precursor to keratin.

c. Stratum granulosum

- It contains 3-5 rows of flattened cells.
- Its mainly composed of 'keratohyalin granules' (darkly stained)

d. Stratum spinosum

- Also known as "spiny layer"
- It contains 8-10 rows of polyhedral (many sided) cells
- It has the appearance of prickly spines.

- It mainly contains melanin granules and Langerhans' cell.

e. Stratum basale

- Also known as "stratum germinativum"
- It's the deepest epidermal layer.
- Its attached to dermis
- Its composed of single row of cells, mostly columnar keratinocytes.

2. Dermis

Dermis lies between the epidermis and subcutaneous layer and contains many type of sensory receptor for touch, pressure, vibration, pain, temperature etc.

The type of cells located in the dermis is:

- Fibroblasts, Mast cells, Histocytes.
- Hair follicles, nerves.
- Lymphatic vessels and blood vessels.
- Sweat glands and oil glands.

Dermis contains two layers:

- a. Papillary layer
- b. Reticular layer



Fig: Microscopic layers of dermis.

a. Papillary layer

It's composed of loose connective tissue with small surface projections called "Dermal papilla".

It contains:

- Capillaries
- Pain receptors
- Touch receptors (Meissner's corpuscles) - Dermal ridges (pattern called fingerprints)

b. Reticular layer

- Its dense irregular in nature.
- It contains collagen fibers and offer strength.
- It holds water to provide hydration.

3. Hypodermis

It's known as subcutaneous tissue or Sub-Q.

It consists of loose connective tissue with collagen and elastic fibers.

Main type of cells in hypodermis is:

- Fibroblast
- Adipose tissue
- Macrophages

Functions as:

- Energy source
- Provides Insulation
- Its common site for injection
- It anchors skin to underlying structures

Skin Relating Problems

1. Dry skin

- Dry skin has a low level of sebum and can be prone to sensitivity. Dry skin refers to roughened or scaly skin that is less flexible than normal and dry to feel.
- The dry skin is the common problem in all age groups, but more in elderly individuals.
- The water content of the stratum corneum play important role for maintaining the normal appearance and texture of skin.
- It looks dull, mainly on the cheeks and around the eyes.

Causes:

- Poor diet, nutritional deficiencies and especially deficiencies of vitamin A and the vitamin B can also contribute to dry skin.
- Harsh soaps, itchy clothing, exposure to sun, cold environment, chemicals and long hot showers.
- If the oil glands do not produce enough lubrication to the skin it may lead to dry skin.
- Medications you take and even medical conditions such as diabetes, hypothyroidism, malnutrition can also cause severe dry skin.

Care for dry skin:

- Skip long hot shower, Hot water strips oils from the skin faster than warm water.
- Use a gentle cleanser or shower gel with moisturizer.
- Go for soap free or mild soap cleansers instead of harsh cleanser.
- Moisturize while skin is moist moisturizer increases the water content of the outer layers of skin and gives it a soft look.

2. Acne

Acne is an inflammatory skin condition that occurs when oil and dead skin cells block the pore opening causing sebum to build up inside the pores.

Causes:

- Genetics is the primary cause of acne in 80% of cases.
- Hormonal activity, such as occurs during menstrual cycles and puberty may contribute to the formation of acne.
- Stress can cause or worsen acne.
- Several medications can worsen pre-existing acne. Examples of such medications include hydantoin, isoniazid, and testosterone.

Treatment:

- Using cleansers for oily skin and toners help to remove excess sebum.
- Avoid using fatty skin care and cosmetic products; use non comedogenic products (designed and proven not to clog the follicles).
- Do not use harsh products or over clean acne-prone skin as this can lead to inflammation.
- Use anti-androgen, anti-biotic like tetracycline, cotrimoxazole to stop growth of bacteria and reduce inflammation.

3. Pigmentation

- Pigmentation refers to the colouring of the skin. Skin pigmentation disorders cause changes to the colour of your skin.
- Skin gets its color from pigment called melanin.
- It occurs due to the deposition of the melanin, which is produced by specialized cells called melanocytes.

Type of pigmentation:

- a. Hyperpigmentation
- b. Hypopigmentation
- a. **Hyperpigmentation** means that the skin looks darker than normal (often brown to grey). This is the more common response after inflammation.
- b. **Hypopigmentation** means that the skin looks lighter than normal.

Causes:

- Eczema
- Psoriasis
- Skin allergies (contact dermatitis)
- Injury To the skin
- Insect bites
- Acne

Treatment:

- Topical prescription medication such as 4% hydroquinone are often use to treat Hyperpigmentation.
- Aloe Vera contain aloin, a natural depigmenting compound.
- Over the counter brightening product that contain licorice root, pine tree, vitamin E, vitamin C.

4. Prickly heat

The condition that we call prickly heat, also known as heat rash, happens to adults and children when sweat becomes trapped under the skin.

Causes: Heat rash is caused by a blockage of the sweat gland, especially after repeated sweating, leading to inflammation which

causes itching, prickling sensation and red rashes, which may be extensive.

Treatment:

- Remain in a cool environment e.g. air conditioning, or close to a fan, and allow for adequate ventilation of the skin.
- Take cool showers or baths several times a day and always after exercise or heat exposure.
- Use cool packs on affected areas

5. Wrinkles

Wrinkles are lines and furrow that form in your skin. These are especially noticeable around your mouth, neck, hands, and eyes.

Causes:

- Ageing
- Repeated facial expression
- Exposure to uv-sun light
- Smoking

Treatment:

- Topical retinoids: Derived from vitamin A, retinoids can be applied to skin to reduce fine wrinkles, splotches and skin roughness.
- Antiwrinkle creams.
- Rhytidectomy: is a type of cosmetic surgery procedure used to give a more youthful facial appearance.

6. Body odour

Body odour is the perceived unpleasant smell our bodies can give. Body odour caused by the combination of sweat and bacteria normally found in skin.

Causes:

- Body odour is caused by bacteria breaking down sweat and is largely linked to the apocrine glands.
- Excessive sweating.

Treatment:

- Keep the armpits clean
- Use deodorant or antiperspirant:
- Have a shower or bath at least once a day
- Clothing: Natural fibers allow your skin to breathe, resulting in better evaporation of sweat.

Structure of hair and Hair growth cycle

Hair is a simple structure that is made up of Protein filaments called Keratin.

The main functions of Hair are:

- Thermo regulation
- Protection
- Facilitation for evaporation of perspiration
- Heat insulation on head

Trichology - Branch of medical science that study hair and its diseases and care.

Hair is made up of two separate structures:

1. **The hair root:** which exists below the skin.
2. **The hair shaft:** which is the hair that we see.

Structure of Hair root

1. **Hair follicle:** tube - like depression or pocket in the skin or scalp that contains the hair root.
2. **Hair bulb:** lowest part of a strand of hair; thickened, club- shaped.
3. **Dermal papilla:** small, cone-shaped elevation located at the base of the hair follicle that fits into the hair bulb.
4. **Arrector pili muscle:** small, involuntary muscle in the base of the hair follicle; when it contracts, we get goose bumps.
5. **Sebaceous glands:** oil glands in the skin that are connected to the hair follicles; secretes sebum (fatty, oily substance)

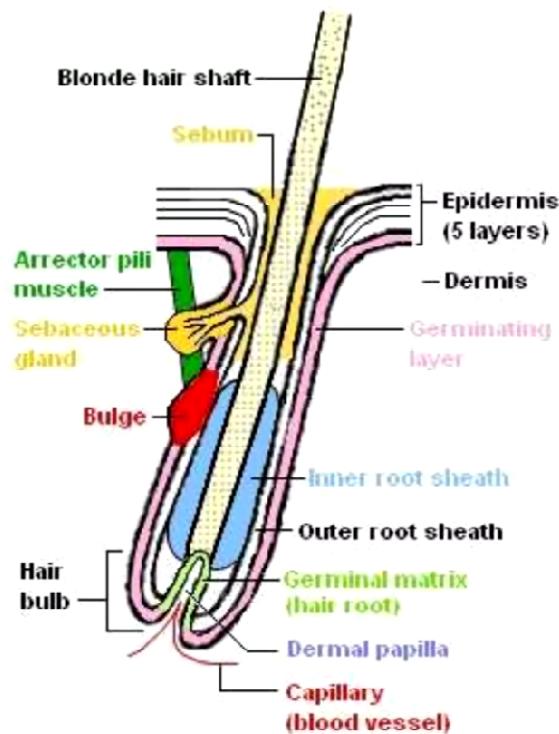


Fig: Hair root structure.

Structure of Hair shaft

The shaft has three layers:

1. Cuticle (outer layer)
2. Cortex (middle layer)
3. Medulla (inner layer)

The medulla is a honeycomb keratin structure with air spaces inside.

The cortex gives flexibility and tensile (stretching) strength. to hair and contains melanin granules, which give hair its colour. its cortex is made from tiny fibers of keratin running parallel to each other along the length of the hair shaft.

The cuticle is made from 6 to 11 layers of overlapping semi-transparent keratin scales (which make the hair waterproof and allow it to be stretched).

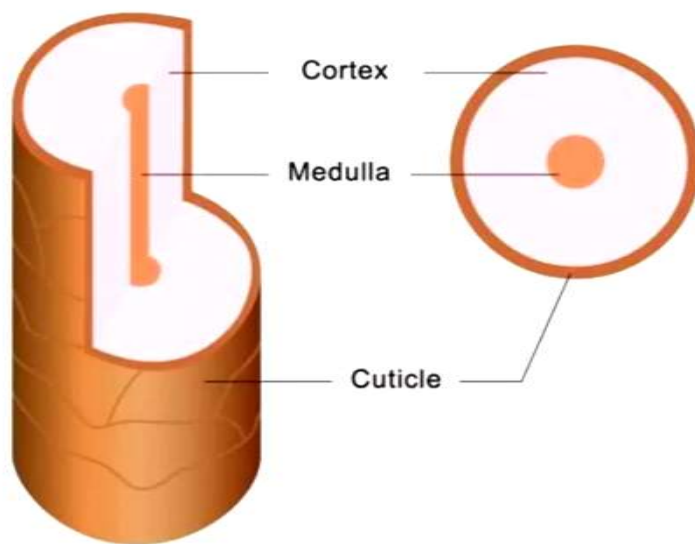


Fig: Layers of hair shaft.

Hair Growth Cycle

The three stages of hair growth are the:

1. Anagen Phase
 2. Catagen Phase
 3. Telogen Phase
- Each strand of hair on the human body is at its own stage of development.
 - Once the cycle is complete, it restarts and a new strand of hair begins to form.
 - The rate or speed of hair growth is about 1.25 centimetres per month, or about 15 centimeters per year.

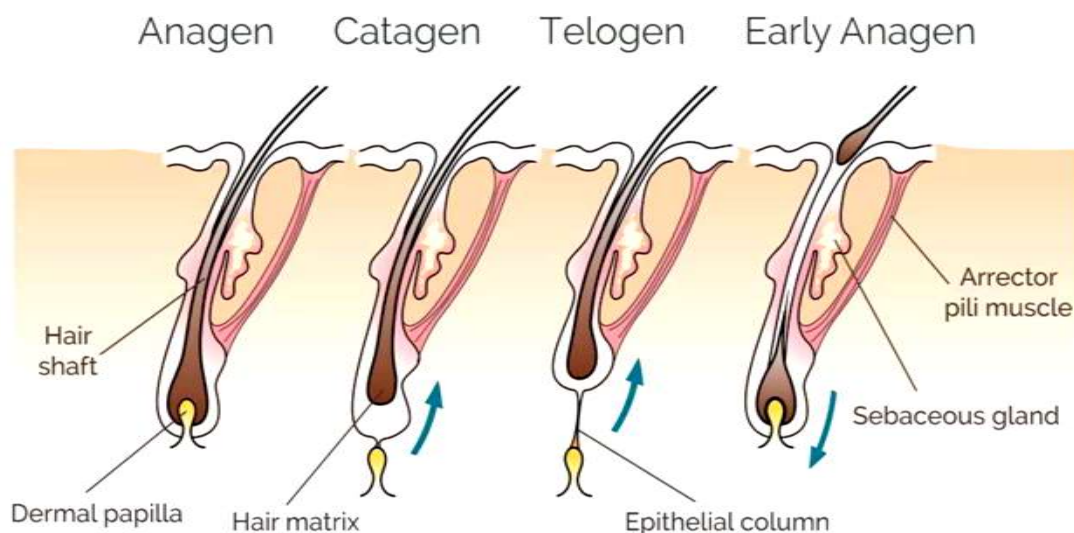


Fig: Hair growth cycle.

1. Anagen

- Anagen is the hair growing phase. Hair growth rate is about 3 mm per week.
- It begins in the papilla and can last from two to six years.
- The longer the hair stays in the anagen phase, the longer it will grow.
- About 85% - 90% of the hairs on one's head are in the anagen phase at any given time

2. Catagen

- The catagen phase is a short transition stage that occurs at the end of the anagen phase.
- This process cuts the hair off from its blood supply and from the cells that produce new hair.
- This phase lasts for about 2-3 weeks and then the hair follicle enters the telogen phase.

3. Telogen

- The telogen phase is the resting phase of the hair follicle.
- Fifty to one-hundred club hair are shed daily from a normal scalp.
- Then, when telogen ends, hair germ cells start growing a new hair follicle from the dermal papilla and the next hair growth cycle starts again.

Common problems associated with oral cavity

1. Bad Breath

- Certain foods, health conditions and habits are the causes of bad breath.
- Bad breath, also called 'Halitosis'.
- In many cases, you can improve bad breath with consistent proper dental hygiene.
- The most common cause of halitosis is bad oral hygiene.
- If particles of food are left in the mouth, their breakdown by bacteria produces sulfur compounds.

Causes

- Tobacco
- Food
- Throat conditions
- Diseases
- Crash diet
- Drugs
- Foreign body
- Dry mouth

Treatment

- Diet
- Avoid dry mouth
- Brush Tounge
- Clean dentures
- Floss
- Brush the teeth

2. Dental Carries (Tooth Decay)

- Dental carries also called as Tooth decay, is a breakdown of teeth due to acids made by bacteria.
- The most common bacteria associated with dental cavities are the mutans streptococci, (Streptococcus mutans, Streptococcus sobrinus, Lactobacilli).

* Causes

- Accumulation of food particles.
- Accumulation of bacteria in gelatinous mass. Called as "plaque".
- Anaerobic fermentation of sugar in saliva.
- Release of lactic acid.
- Breakdown of minerals present in enamel.
- Affect internal compound-dentin, pulp tissue.

* Precautions

- Low sugar containing diet.
- Low fluoride.
- Proper brushing

■ Treatments

- Fluoride treatment
- Filling
- Crown
- Root canal

3. Periodontal diseases

- Periodontal disease is a serious gum infection that damages the soft tissue and destroys the bone that support the teeth.
- Periodontal disease can cause teeth to loosen or lead to tooth loss.
- The most common bacteria associated are, (porphyromonas, straptococcus, Actinomyces).

Types

1. Gingivitis

- Gingivitis is a non-destructive type of periodontal disease.
- Have signs of some inflammation of the gingival tissue at the necks of the teeth.

2. Periodontitis

- Untreated gingivitis can progress to periodontitis.
- Characterised by the formation of pockets or spaces between the tooth and gums.

Symptoms:

- Swollen or puffy gums
- Bright red, dusky red or purplish gums
- Gums that feel tender when touched
- Gums that bleed easily
- Gums that pull away from your teeth (recede), making your teeth look longer than normal
- New spaces developing between your teeth
- Painful chewing

Prevention/Treatment:

- Proper hygiene of teeth
- Brushing properly on a regular basis
- Regular dental check-ups and professional teeth cleaning as required.

Drugs:

- Tetracyclin
- Deoxycyclin
- Macrolite

4. Stomatitis

Stomatitis is an inflammation of the mucous lining of the mouth, which may involve the cheeks, gums, tongue, lips, and roof or floor of the mouth. The word "stomatitis" literally means inflammation of the mouth.

Types:

- Nutritional deficiency stomatitis
- Aphthous stomatitis
- Angular stomatitis
- Denture-related stomatitis
- Allergic contact stomatitis
- Migratory stomatitis
- Herpetic gingivostomatitis

* Causes

- Chemotherapy
- Radiotherapy
- Loose-fitting dental prosthetics
- Trauma
- Poor dental hygiene
- Smoking
- Dehydration
- Medication
- Burns

* Symptoms

- Pain or discomfort in the mouth.
- The presence of open sores or ulcers in the mouth.
- Fever ,sometimes as high as 101–104°F .
- Irritability and restlessness
- Blisters in the mouth
- Swollen gums , which may be irritated and bleed.
- Drooling.
- Dysphagia.
- Foul-smelling breath.

Treatments

Medical management:

- Bismuth salicylate, sucralfate, antacids
- Water -Soluble lubricants from mouth and lips
- Topical analgesics, such as benzamine hydrochloride
- Topical anesthetics, such as lidocaine viscous
- Oral or parenteral analgesics, including opioids if needed, for pain not controlled with above

Other management:

- Antiseptic mouth wash
- Avoid excessive brushing
- Denture hygiene measures

5. Dry mouth

- It cause due to the decrease secretion of saliva.
- Dry mouth is often due to the side effect of certain medications or aging issues or as a result of radiation therapy for cancer.

* Symptoms

- Dryness or a feeling of stickiness in your mouth
- Saliva that seems thick and stringy
- Bad breath
- Difficulty chewing, speaking and swallowing
- Dry or sore throat and hoarseness
- Dry or grooved tongue
- A changed sense of taste

* Causes

- Medications
- Aging
- Cancer therapy
- Nerve damage
- Other health conditions
- Tobacco and alcohol use

Treatment

- 1. Change medications that cause dry mouth** - If your doctor believes medication to be the cause, he or she may adjust your dosage or switch you to another medication that doesn't cause a dry mouth.
- 2. Recommend products to moisturize your mouth** - These can include prescription or over-the-counter mouth rinses, artificial saliva or moisturizers to lubricate your mouth. Mouthwashes designed for dry mouth, especially ones with xylitol, can be effective, such as Biotene Dry Mouth Oral Rinse or Act Dry Mouth Mouthwash, which also offer protection against tooth decay.

6. Tooth Erosion

- Erosion is the wearing away of the tooth surface by an acid, which dissolves the enamel and the dentine. There are a variety of ways that tooth structure is lost.
- Food and beverages can dissolve tooth structure if they are acidic enough.
- Two sources of acid in the mouth are dietary and gastric, and the acid dissolves the calcium in the tooth.
- Beverages with high acid content include all carbonated beverages and citrus-based drinks. Additionally, wine has been shown to erode teeth, with the pH of wine as low as 3.0-3.8.

* Causes

- Dietary
- Regurgitation
- Environmental
- Flow of saliva
- Exposed dentin
- Idiopathic

▪ Sources of Dietary Tooth Erosion

- Citric acid in soft drinks
- Acidic fruit juices
- Acidic sugar-free drinks
- Fruits

* Treatments

- Modifying the pH of the food or beverage contributing to the problem
- Changing the patient's lifestyle to avoid the food or beverage.
- Decreasing abrasive forces.
- Using a re-mineralizing agent (sodium fluoride solution) Applying fluoride gels or varnishes to the teeth.
- Drinking milk or using other dairy products.
- Using a neutralizing agent such as antacid tablets.

7. Cavities

- A cavity, also called tooth decay, is a hole that forms in your tooth.
- Cavities start small and gradually become bigger when they're left untreated. Because many cavities don't cause pain in the beginning, it can be hard to realize that a problem exists.
- Regular dental appointments can detect tooth decay early.

* Symptoms

- tooth sensitivity
- tooth pain
- a visible hole in your teeth
- black or white staining on your teeth

■ Causes

- bacteria
- saliva
- acid
- food particles

* Treatments

- Tooth fillings
- Crowns
- Root canal

Prevention

- a. Brush your teeth at least twice per day with a fluoride toothpaste.
- b. Floss at least once daily, as recommended by the American Dental Association.
- c. Eat fewer sugary and acidic foods, like sweets, candy, juice, soda, and refined carbohydrates.
- d. Limit snacking between meals.
- e. Consider getting dental sealants on your teeth.

Cleansing and Care Need for Face, Eye Lids, Lips, Hands, Feet, Nail, Scalp, Neck, Body And Under Arm.

Skin care products

1. Face Wash

- Is a facial care cosmetic used to cleanse the skin.
- Helps to improve skin complexion.
- Helps rejuvenates the skin.
- Helps to control excessive oil.
- Helps to nourish the skin and make the bright and radiant.

Examples: -

1. Papaya rehydrating face wash
2. Neem tulsi face wash
3. Sandal almond face wash

2. Cleanser

- A cleanser is facial care product that is used to remove makeup, dead skin cells, oil, dirt, and other type of pollutants from the skin of the face.
- This helps to unclog pores and prevent skin condition such as acne.
- A cleanser can be used as part of a skin care regimen together with a toner and moisturizer
Eg: charcoal cleanser.
- It is an exclusive skin lightening formula.
- Super absorbency and anti bacterial activity make an ideal ingredient for oily and acne pore skin type
- It acts like magnet to attract and absorb dirt.

3. Moisturizer

- It is a cosmetic preparation used to prevent dryness in the skin.
- If not moisturized properly, skin tends to look dull, dry, flaky, might even cause itching.

Moisturising products are classified into ;

- Day preparations
- Night preparations
- Hand and body lotions
- All purpose products
- Barrier creams

4. Fairness Cream

- Also known as whiteners, skin brighteners, or skin lighteners cream.
- They work by reducing the pigment called melanin in the skin.
- This is the most common technique to naturally lighten darkened skin.
- Helps visibly lighten the hyper pigmentation in the skin, nourishes skin to bright.

Lips Care Products

Care needs for lips:

- Do not touch or lick your lip.
- Follow healthy diet
- Stay hydrated by drinking lot of water.
- Keeps your lips hydrated over night.
- Massage your lips.
- Scrub your lip
- Always carry a lip balm
- Use lipstick when stepping out.

1. Lip Balms

- Lip balm is a wax-like substance you apply to moisturize and ease pain caused by chapped or dry lips.
- Due to thin skin, lips are vulnerable and usually display signs of dryness before other areas of the skin.
- Lip balm helps protect lips from dry air, wind, and cold temperatures.
- Lip balms come in a variety of different flavors and pleasant scents.
- Different to other parts of our skin, lips don't have sebaceous or sweat glands, and are covered only with a thin skin
- Due to bad and variable weather conditions (wind, coldness, dry air, sun, UV radiation...), our lips very easily lose their moisture and softness.

Advantages

- a. Skin heals faster
- b. Avoid the embarrassment of chapped lips. • Contains SPF protection too

Disadvantages

- a. Lip rash (if brand contains allergens)
- b. Some brands contain dangerous ingredient
- c. There is a risk of possible infection. Continuous use of a product may lead to contamination.

2. Lip gloss

- Lip gloss is a product used primarily to give lips a glossy lustre, and sometimes to add a subtle color.
- It is distributed as a liquid or a soft solid (not to be confused with lip balm, which generally has medical or soothing purposes) or lipstick, which generally is a solid, cream like substance that gives off a more pigmented color.

Contents:-

Like lipstick, lip gloss is a mixture of waxes, oils, and pigments. However, lip gloss contains less pigments, and those used are often pale in color or diluted (<3%). Furthermore, the free-flowing nature of the product requires less wax. The principal components are lanolin, which feels good on the lips due to its moisturizing qualities and imparts gloss, and polybutene.

3. Lip Liner:-

- Lip liner, also known as a lip pencil, is a cosmetic product.
- It is intended to fill in uneven areas on the outer edges of the lips before applying lipstick to give a smoother shape.
- Alternatively, lip liner can be used to fill in the entire lip before the application of lipstick, and in some cases is worn as a lipstick on its own.
- The product is usually sold in a retractable tube or pencil form which can be sharpened.
- Lip liner is usually available in the same range of colors as lipsticks: e.g., reds, pinks, browns, plums, etc.
- Lip liner also comes in invisible, for giving the illusion of smooth lips without adding or affecting

Ingredients - Like lipstick, lip liners are composed of waxes, oils, and pigment.

4. Lipstic

- Cosmetic products containing oils, waxes, pigments and emollient that apply color, texture and protection to lips
- These preparation enhances the appearance of lip and are non tacky.
- They are made from hydrophobic material.
- When the solid formulates to the lip surface friction melts its briefly and allow for transfer.
- The materials cools and reforms creating a film that sticks to surface due to hydrophobic interactions.

Care Preparations For Eyelid

1. Eye Shadow

- Eye shadows are used to give a background of color to the eye.
- They are marketed in various shades
- Conventional shades are blue, green, and brown.
- They may be either in solid form or liquid paste form. • Solid form eye shadow may be powder compact type, oil based stick type or pencil type.
- Liquid base form of eye shadows may be oil base paste or emulsion type: - o/w or w/o emulsion type.

2. Mascara

- Mascara is a cosmetic commonly used to enhance the eyelashes.
- It may darken, thicken, lengthen, and/or define the eyelashes.
- Normally in one of three forms-liquid, cake, or cream-the modern mascara product has various formulas; however, most contain the same basic components of pigments, oils, waxes, and preservatives.

Different types of mascara

- a. Water soluble Mascaras.
- b. Water proof Mascaras.

3. Eye liner

Eye liner or eyeliner is a cosmetic used to define the eyes. It is applied around the contours of the Eye.

Types

- a. **Liquid Eyeliner** - In liquid form. Applied with help of brush.
- b. **Gel Eyeliner** - Gel eyeliner comes in a pot and is applied with a small brush. Gel can also be used to create bold lines and catlike eyes.

- c. **Eyeliners Pencil (Kohl)** - Comes as a pencil and applied directly to the eyelids to create a variety of aesthetic effects.

Care for Hands

- Wash with moisturizing hand soap.
- Clean under your nails with a nail brush.
- Keep your nails trim and well shaped.
- Exfoliate your hands weekly.
- Treat your hands with a mask weekly.
- Apply a retinol treatment to dark spots.

1. Hand Moisturizer

- Keeping your hands moisturized can turn back the clock and there are even combined products that also keep nails healthy too as an added bonus.
- Moisturizing hands is especially important in winter time when harsh weather can leave skin dry and even cracked.

2. Hand Sanitizer

- A hand sanitizer or hand antiseptic is a non-water-based hand hygiene agent.
- Most are based on isopropyl alcohol or ethanol formulated together with a thickening agent such as Carbomer into a gel, or a humectant such as glycerine into a liquid, or foam for ease of use and to decrease the drying effect of the alcohol.
- Hand sanitizers containing a minimum of 60 to 95% alcohol are efficient germ killers.
- Alcohol rub sanitizers kill bacteria, multi-drug resistant bacteria (MRSA and VRE), tuberculosis, and some viruses (including HIV, herpes, RSV, rhinovirus, influenza, and hepatitis) and fungi.

- Hand sanitizers are most effective against bacteria and less effective against some viruses. Alcohol based hand sanitizers are almost entirely ineffective against Norwalk type viruses, the most common cause of contagious gastroenteritis.

Feet Care

1. Washing
2. Keeping Them Dry
3. Moisturising
4. Removing Jagged Skin
5. Wearing Socks
6. Wearing Comfortable Shoes
7. Apply Anti Fungal Foot Creams

Care for Neck

We shouldn't neglect our necks, as they show signs of aging even more quickly than our faces do and between wrinkling, sagging and sun damage (not to mention the ever-present fear of "tech neck"), we've been thinking a lot more about how to care for this essential part of the body.

So we went straight to the source and asked a dermatologist how we can best protect our necks

Neck care cosmetics

1. Moisturising cream
2. Cleansing cream
3. Face wash
4. Fairness cream

Care for Scalp

1. Protect your scalp from the sun.
2. Shampoo Regularly.
3. Limit Chemical Treatment.
4. Massage.
5. Preserve Moisture.

Care preparation for scalp

- Shampoos
- Hair Oils

Care Preparation for Nail

Nail bleachers and stain removers

- They are used to whiten the nails and to remove discolourations.
- These preparations were popular before introduction of nail lacquers.

Formulation:

Nail bleachers are generally oxidizing agent like hydrogen peroxide, sodium perborates.

- Zinc peroxide 7%
- Talc 23%
- Titanium dioxide 20%
- Petroleum jelly 25%
- Mineral oil 25%

1. Nail lacqures

Nail lacqures or nail enamers are the one of the group of nail cosmetics.

A nail lacqures should have the following properties:-

- It should be harmless to skin and nail
- It should convenient and easy to apply
- It should stable on storage
- It should form satisfactory film on nail formulation .

Formula

- methylated spirit 10%
- Ethyl acetate 20%
- Butyl acetate 15%
- Toluene 55%

Care for Under Arm

1. Hair Removal

Shaving: Shaving the right way will eliminate stubble while preventing ingrown hairs and any possible inflammation. Remember that hair under there can sometimes grow in various directions, which means you may have to shave in more than just one downward motion. Gently exfoliate the area prior to shaving (your loofah will do or use The Body Shop Aloe Gentle Exfoliator) and use a conditioning shave cream for smooth results. (Exfoliation also helps fight off underarm odour, but more on that later!) For a close and comfortable shave, be sure to use a fresh blade.

Waxing: While waxing isn't the most fun, it's highly effective and can leave you hairless for weeks instead of days. Try it at home if you're

brave (and not the messy type!) or seek the help of a professional esthetician or wax expert to ditch the daily stubble.

Deodorant

- A deodorant is a substance applied to the body to prevent body odour caused by the bacterial breakdown of perspiration in armpits, feet and other areas of the body. A subgroup of deodorants, antiperspirants, affect odour as well as prevent sweating by affecting sweat glands. Antiperspirants are typically applied to the underarms, while deodorants may also be used on feet and other areas in the form of body sprays.
- Deodorants are often alcohol-based. Alcohol initially stimulates sweating, but may also temporarily kill bacteria.
- Deodorants can be formulated with other, more persistent antimicrobials such as triclosan that slow bacterial growth or with metal chelant compounds such as EDTA. Deodorants may contain perfume fragrances or natural essential oils intended to mask the odour of perspiration. In the past, deodorants included chemicals such as zinc oxide, acids, ammonium chloride, sodium bicarbonate and formaldehyde, but some of these ingredients were messy, irritating to the skin or even carcinogenic.

Formula: -

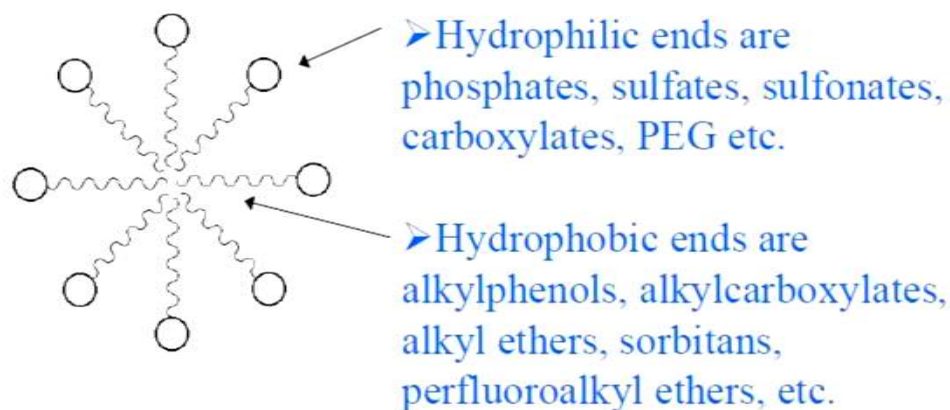
- Water
- Alcohol
- Dimethicone
- Propylene glycol
- PEG
- Fragrance (flower oil)

Unit 3 - Formulations and Building Blocks

Surfactant

- A substance which tends to reduce the surface tension of a liquid in which it is dissolved.
- Surfactant is amphiphilic in nature.
- Amphiphilic - having both hydrophilic & hydrophobic parts.
- Hydrophilic - lipophile (polar, high affinity)
- Hydrophobe - lipophile (non polar, alkyl chain)

Structure of Surfactants in Aqueous Solution



- The primary performance function surfactant is the lowering of interfacial tension.
- Ex. Cleansing, emulsion, dispersion
- Cleansing-lowering tension between the oil / dirt, water and your skin or hair
- Emulsion- lowering tension between oil phase and water phase
- Dispersion- between particles and water / oil

Classification

1. Anionic surfactants:- Sulfate, sulfonate
2. Nonionic surfactants:- fatty acid ester, ethoxylate
3. Cationic surfactants :-quaternary ammoniums
4. Zwitterionic surfactants:- amino acid derivatives

1. Anionic surfactants

- Anionic surfactants work best to remove dirt, clay, and some oily stains.
- These surfactants work to follow ionization.
- When added to water, the anionic surfactants ionize and have a negative charge.
- The negatively charged surfactants bind to positively charged particles like clay.
- Anionic surfactants are effective in removing particulate soils.
- Anionic surfactants play a major role in modern cleaning products due to their superior detergency performance. Example:
- Sulfonate, phosphate, sulfate and carboxylates. Alkyl sulfates include ammonium lauryl sulfate, sodium lauryl and the related alkyl-ether sulfates sodium laureth sulfate, also known as sodium lauryl ether sulfate (SLES), and sodium myreth sulfate.

2. Nonionic surfactants

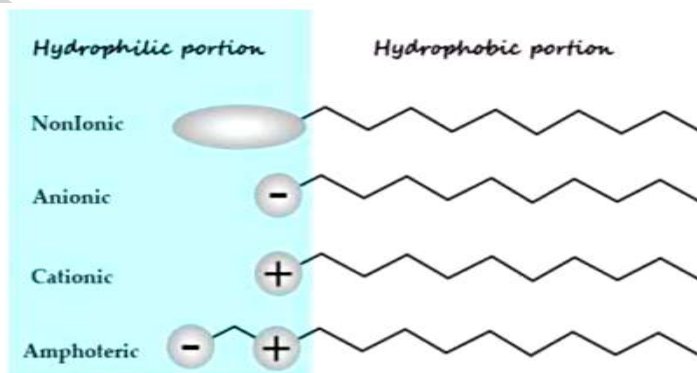
- It Have no foaming capabilities so this group of surfactants are rarely used.
- These surfactants do not have a charge in their water- loving heads.
- Nonionic surfactants (like Polysorbate 20) perfect solubilizers.
- Some common nonionic surfactants are Polysorbates, Emulsifying Wax NF, E-wax, Glyceryl Oleate, Glyceryl Stearate, ingredients with the prefix PEG, Cetareths, Oleths, Sorbitans, Lauryl Glucoside, and Polyglycose.

3. Cationic Surfactants

- Cationic surfactants are the opposite of anionic surfactants. They have a positively charged water-loving head.
- It is because of this positive charge that cationic surfactants can offer many skin loving, nourishing benefits to the skin and body.
- Cationic surfactants alone do not allow for foaming capabilities.
- Some common cationic surfactants are Chlorides (Benzalkonium, Stearalkonium, and Centrimonium), Trimethyl Ammoniums, and Methyl Sulfates.

4. Zwitterionic (amphoteric) surfactant

They have both cationic and anionic centers attached to the same molecule.



	Anionic surfactants	Cationic surfactant	Zwitterionic surfactants	Nonionic surfactant
Surfactant ion	Negatively charged	Positively charged	Both positive and negative parts	No ionization
Examples	sulfonic acid salts, alcohol sulfates, alkylbenzene sulfonates, phosphoric acid esters, and carboxylic acid salts	polyamines and their salts, quaternary ammonium salts, and amine oxides	β -N-Alkylaminopropionic acids, N-alkyl- β -iminodipropionic acids, N-alkylbetains, sulfobetaines, sultaines	polyoxyethylenated alkylphenols, alcohol ethoxylates, alkylphenol ethoxylates, and alkanolamides
Toxicity	Relatively nontoxic	toxic	Relatively nontoxic	Relatively nontoxic
Sorption to soil	No sorption	Strong sorption	Can be adsorbed	Not significant sorption
Application in environmental area	Good solubilizer, widely used in petroleum oil recovery, contaminant hydrogeology remediation	Not widely used in environmental application	Compatible with other surfactants and therefore can be mixed together as a cosurfactant in petroleum and environmental application	Good solubilizer, can be used as cosurfactant in petroleum and environmental application

Emollients

- Meaning soother or softener, an emollient softens dry, rough, flakey skin, making it look and feel better.
- Emollients are cosmetic preparations used for protecting, moisturizing, and lubricating the skin.
- Emollients that contain a lot of oil are also occlusive agents. This means they coat your skin with a thin oily film that seals the water in your skin. Occlusive agents keep your skin hydrated longer.
- The most common way to classify emollients is by how oily or greasy they are.
- All emollients soften and smooth your skin, but the amount of oil in them determines how strong of an occlusive agent they are.
- The more oil an emollient contains, the better it is at forming a protective layer on your skin to trap moisture.

Types of emollients

- Emollients may be medicated or non-medicated.
- Non-medicated topical moisturizers contains occlusive agents, emollients, and humectants.
- Occlusive agents coat the skin, forming a physical barrier that prevents the loss of water.
- Emollients an occlusive barrier and they smooth flaky skin cells, to make the skin look smoother. Some spread more easily than others. Esters and oils can be used.
- Humectants hey attract water from the atmosphere and from the lower layers of skin to moisturize the surface of the skin.

Uses

Indications	Examples
Eczemas ^[1,3,4]	Atopic dermatitis Hand eczema Nummular eczema Stasis eczema Asteatotic eczema
Disorders of keratinization ^[70-72]	Ichthyosis Ichthyosis vulgaris Bullous ichthyosis erythroderma Lamellar ichthyosis Acquired: Drugs and malignancies Psoriasis Keratosis pilaris Palmoplantar keratodermas Lichen planus
Dermatosis secondary to an underlying disorder ^[1]	Diabetes Thyroid disorders Pruritus of pregnancy Uremic pruritus Pruritus secondary to hepatic disorder Exfoliative dermatitis Chemotherapy-induced xerosis
Other dermatologic disorders ^[73,74]	Lichen simplex chronicus Acne vulgaris Rosacea Retinoid-induced irritant dermatitis
Maintenance of Skin integrity in special populations ^[75-77]	Elderly patients Neonates and infants Amputee and bed ridden patients
Miscellaneous uses	Xerosis ^[71] Contact dermatitis ^[30] Anti-aging ^[27,28]

Rheological Additives

- Rheology can be defined as "the science or study of how things flow"
- Rheological modifiers are often referred to as thickeners, and whilst increasing the apparent viscosity will confer a feeling of "quality" to the formulation, this is only one aspect of rheological control.
- The product itself can be Newtonian or pseudoplastic, thixotropic, be a ringing gel or a stringy flowable liquid.
- Different types of rheology modifier are available to the formulator. These include natural gums such as guar and starch, modified naturals such as cellulose derivatives, synthetics such as acrylic polymers and inorganic such as clays.

Classifications: -

1. Polymeric Rheology Modifiers: These, usually acrylate based polymer thickeners, are cost effective and efficient at low use levels, can provide suspension of particles but can be sensitive to salt content and tend to short rheology.

2. Mineral Colloidal Modifiers:

- Minerals (naturally sourced) such as Magnesium Aluminum Silicate, Bentonite, and Hectorite can be used to create colloidal systems which impart viscosity in a non-Newtonian manner.
- Typically synergistic when used with gums (xanthan). They also have a smooth, 'dry' feel.
- Hydrophobically modified minerals can also provide rheology modification in non-aqueous systems.

3. Associative Thickeners: These thickeners interact with surfactants in a way that moderates flow and adds viscosity. They can be sensitive to different ingredients including sulfate free surfactants.

4. Cellulosic Thickeners: Based on cellulose (wood pulp), these synthetically modified polymers are similar to the polymeric thickeners in that they can be highly efficient. At higher use levels they can feel slippery or stringy and do not provide suspending properties. Hydrophobic modification can allow for use in solvent systems

Antimicrobial used as preservative

Introduction:

Preservative is a substance or chemical that is added to cosmetic product to prevent decomposition by microbial growth or by undesirable chemical changes.

These are antimicrobial agents which maintains the microbiological safety of cosmetic products by inhibiting growth of microbes.

Commonly used preservatives:-

1. Parabens.
2. Formaldehyde releasers.
3. Isothiazolinones.
4. Phenoxyethanol.
5. Organic Acids.

Need of Preservatives:

- To prevent spoilage of cosmetic products.
- There are two main reasons for spoilage of cosmetics:
 1. Oxidation of oils and fats.
 2. Microbial growth.
- To stabilize the product.
- To increase the shelf life of product.
- Water promotes growth of micro-organisms so there is need of preservative in anhydrous cosmetic preparation.

Ideal Characteristics:

- a. Compatible with drug components
- b. Effective
- c. Stable
- d. Odorless
- e. Cost effective
- f. Tasteless
- g. Non-toxic and non reactive
- h. Highly soluble

Merits

- a. It should prevent the bacterial growth ,spoilage.
- b. Keep the product fresh for longer time.
- c. Something that improves appearance of cosmetic product.
- d. Combination of esters exhibit increased activity.

Demerits

- a. Can be toxic in larger concentration.
- b. Sometime interaction with formulation.
- c. Incompatible with some proteins.
- d. Some non-ionic emulsifiers reduce the activity.

Factors affecting effectiveness:-

1. Dissociation and pH
2. Concentration
3. Susceptibility of organisms
4. Interference by ingredients of cosmetics.
5. Influence of solid particles

1. Dissociation and pH:-

- Different cosmetics have different pH and different microbes grow at different pH.
- Majority of preservatives are active in acidic range than in alkaline pH.
- pH influences preservative efficacy.
- Weak acid's activity depend on undissociated amount and undissociation depends on dissociation constant and pH.
Example - Methyl parabenzoic acid is approx. 50% dissociated at pH 8.5, so efficacy is reduced by almost 50% at pH 8.5.

2. Concentration:-

- Some materials used in cosmetic have antimicrobial action, so concentration required is less.
- Some ingredients promotes growth of microbes, so concentration required is more.
- Effective concentration range is from 0.001% (organic mercury comp.) to 0.5-1% (weak acids).
- If combination of preservative used, concentration required is less

3. Susceptibility of organisms:-

Different antimicrobials have different activity against microbes.

Examples:

- Hydroxy benzoates active against moulds, fungi, yeast and less active against bacteria.
- Chlorbutanol has antibacterial and anti-fungal activity.

4. Interference by ingredients:-

- Non-ionic surfactant like Tween 80, PEG 1000, PEG 400 interfere with activity of preservative by exerting protective effect on micro-organism.
- Soaps and anionic surfactants in lower concentrations promote growth of gram negative bacteria.
- Hydrophilic polymer like PEG, gum tragacanth, carboxy-methyl-cellulose and methyl cellulose have reducing efficiency of anti-microbial.

5. Influence of solid particles:-

- Some cosmetics containing insoluble solid like kaolin, chalk, zinc oxide, talc.
- Preservative can be absorbed on the surface of the solid particles.
- Adsorption depend on nature of solid particles.

Building Blocks for Different Product Formulations of Cosmetics

1. Creams

Creams are defined as a semisolid dosage form containing one or more drug substances dissolved or dispersed in a suitable base.

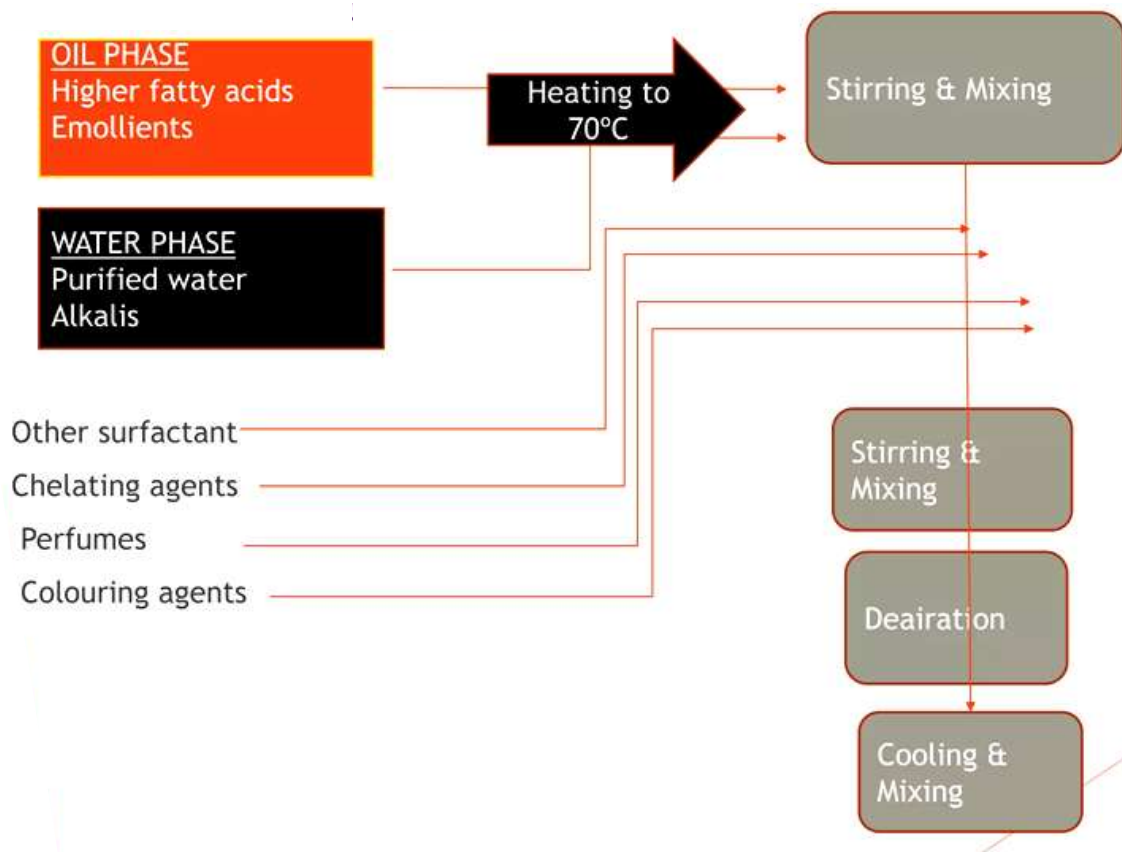
Types of Skin Creams

1. Make up creams
 - a. Vanishing cream
 - b. Foundation cream
2. Cleansing creams
3. Creams for winter
 - a. Cold cream
4. Creams for dry skin
 - a. Moisturizing cream
5. All purpose creams
6. Night creams
7. Skin protective & hand creams

Ingredients used in skin creams

1. Water
2. Oils, fats & waxes
 - a. Mineral oil (Light liquid paraffin, Heavy liquid paraffin)
 - b. Glyceride oil (Almond oil, Arachis oil, Castor oil)
 - c. Fats (Lauric, Palmitic, Stearic acids & Sesame oil, Olive oil, Coca butter, Peanut oil)
 - d. Waxes (Carnauba wax, Beeswax)
3. Lanolin (Derived from wool fat)
4. Glycol (Ethylene glycol, Diethylene glycol & Propylene glycol)
5. Colours (Saffron, Chlorophyll, Cochineal)
6. Emollients :
 - Also known as Moisturizers.
 - Eg: Mineral oil, Squalene, Lanolin
7. Emulsifying agents
 - a. Inorganic solids : Eg: Bentonite, Kaolin
 - b. Gums & Proteins
Eg: Gum Arabic, Gum tragacanth, Gelatin, Egg albumin
8. Wetting agent :
 - Type of surface active agents & lower surface tension.
Eg: Soaps, Sulphonated oils, Fatty alcohol sulphates, Sulphated fatty esters & amides
9. Humectants (Proteins, acids & Polysaccharides)
10. Perfumes (White blossoms, Rosy dreams, Orange blossom)
11. Some functional raw materials like vitamins, amino acids and anti-inflammatory agents may also be incorporated in creams

Manufacture of Creams



1. Moisturizing Cream

- When water is lost from stratum corneum more rapidly than it is received from lower layers of skin the skin becomes dehydrated.
- The dehydrated skin loses its flexibility and appears rough.
- Creams which restore water or moisture & plasticizes the stratum corneum, provides its flexibility and makes it soft. These types of cream are known as moisturizing cream.

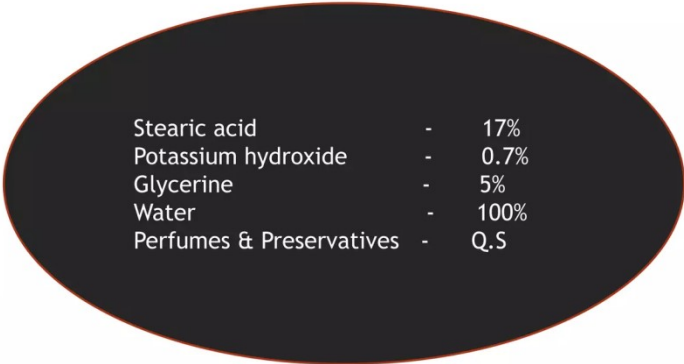
Formula: -

Stearic acid	- 4%
Mineral oil	- 8%
Lanolin	- 1%
Glyceryl monostearate (self-emulsifying)	- 3%
Isopropyl myristate	- 2%
Glycerine	- 4%
Propylene Glycol	- 4%
Triethanolamine	- 0.2%
Water	- 100%
Perfume & Preservatives	- Q.S

2. Vanishing Cream

- Creams which spread easily & seem to disappear rapidly when rubbed on the skin are termed as vanishing creams.
- These creams are composed of emollient esters which leave little apparent film on the skin.
- Traditional formulae of vanishing creams are based on stearic acid.
- Stearic acid melts above body temperature & crystallizes in a form so as to be invisible providing a non-greasy film. It also imparts attractive appearance to the cream.

Formula: -



Stearic acid	-	17%
Potassium hydroxide	-	0.7%
Glycerine	-	5%
Water	-	100%
Perfumes & Preservatives	-	Q.S

Procedure for preparation of vanishing cream :-

- Melt stearic acid in a china dish on a water bath.
- In a beaker, dissolve potassium hydroxide in water & glycerine to it. Heat the aqueous solution upto 70°C on water bath.
- When both aqueous & oily phase reaches the same temperature (70°C), add aqueous phase to melted stearic acid with constant stirring.

- Remove the china dish from heat & continue the stirring, when the temperature reaches 40°C , add perfume and mix uniformly until it becomes cool and a homogeneous cream is obtained. 13

3. Cold Cream

- Cold cream is a w/o type emulsion which when applied to skin, a cooling effect is produced, due to slow evaporation of water present in the emulsion.
- Cold cream is an emulsion of fats and water which can be used to clean & soften the skin.
- Cold cream has been used to remove makeup gently at the end of day & it can also be used to soften tough skin on the knees and elbows or to keep skin protected from harsh winter weather.
- Cold cream is prepared by saponification reaction between beeswax & alkali borax.

Formula: -

White beeswax	-	10g
Liquid paraffin	-	30g
Borax	-	0.5g
Rose oil	-	0.1ml
Water	-	10ml

Procedure for preparation of cold cream: -

- Weigh the required quantity of white beeswax & liquid paraffin and melt in a china dish by heating on a water bath upto 70°C .
- In a glass beaker, dissolve borax in water and heat upto 70°C .

- When both oily and aqueous phases reaches the same temperature (70°C), gradually add borax solution to the melted beeswax drop by drop with constant stirring.
- Stir continuously until it becomes cool. When the temperature lowers to 40-45°C, incorporate rose oil and mix uniformly, until a homogeneous semi-solid mass is obtained.

Evaluation of creams

- 1. Presence of foreign particles / grittiness:** A Small amount of cream was taken and spread on a glass slide free from grease and was observed against diffused light to check for the presence of foreign particles.
- 2. pH of the cream :** About 1g of the cream was weighed & dissolved in 100ml of distilled water and stored for 2 hrs. pH of different formulations was determined by using digital pH meter.
- 3. Viscosity:** Viscosity of the formulation was determined by brook field viscometer using spindle no. S-64 at 20rpm at a temperature of 25°C & determinations were carried out in triplicate and the average of three recordings was recorded.
- 4. Spread ability:**
 - Two sets of glass slides of standard dimensions were taken.
 - The cream formulation was placed over one of the slides.
 - The other slides were placed on the top of formulation, such that the cream between two sides was pressed uniformly to form a thin layer.
 - The spreadability was expressed in terms of time taken by 2 slides to slip off from the cream.
 - Lesser the time taken for separation of 2 slides, better the spreadability.

5. Irritancy test: Mark an area(1 square cm) on the left hand dorsal surface. The cream was applied to the specified area and time was noted. Irritancy, erythema, edema was checked if any, for regular intervals upto 24hrs & reported.

6. Patch test: About 1-3g of cosmetic to be tested was applied on to the sensitive part of the skin. Control patches were also applied. The site of patch was inspected after 24 hrs.18

7. Stability studies:

- a. **Globule size:** 1ml of cream was diluted to 10ml with glycerine. A few drops of this were transferred onto a glass slide & was focussed in a microscope. By using eyepiece micrometer, the diameters of 200 particles were determined randomly.
- b. **Phase separation:** The formulated cream was kept intact in a closed container at 25-30°C not exposed to light. Phase separation was observed carefully every 24hrs for 30 days. Any change in phase separation was recorded.

8. Partition coefficient of the cream:

- 50mg of cream was taken in a separating funnel containing 1:1 ratio of buffer 7.4 & n-hexane.
- Then, solution was shaken occasionally & both phases were separated and filtered & the amount solubilized in each phase was determined by measuring the absorbance using UV spectrophotometer.

9. Extrudability:

- The formulations were filled in a standard collapsible capped tube and sealed.

- The tube was weighed & recorded. The tube was placed between two glass slides & was clamped.
- A 500g weight was placed over the slide & cap was opened.
- The amount of cream were collected & weighed.
- The amount of cream extruded was calculated and grades were allotted.

10. Thermal stability:

- The formulated cream was inserted into glass bottle with the help of a spatula, and tapped to settle to the bottom, filled upto 2/3rd capacity of bottle & insert plug and tighten the cap.
- Filled bottle was kept erect inside the incubator at $45 \pm 1^\circ\text{C}$ for 48hrs.
- The sample will pass the test, if on removal from the incubator shows no oil separation or any other phase separation.

4. Shampoo

A Shampoo is a preparation of a surfactant in a suitable form - liquid, solid or powder which when used under the specialized conditions will remove surface grease, dirt and skin debris from the hair shaft and scalp without adversely affecting the user.

Ingredients for shampoo

1. Surfactants:

a. Anionic surfactants:

- Alkyl benzene sulphonates
- Alpha olefin sulphonates
- Sulphosuccinates

b. Non-ionic surfactants

- Fatty acid alkanolamides
- Polyalkoxylated derivatives
- Amine oxides

c. Amphoteric surfactants :

- N-alkyl aminoacids
- Betains
- Alkyl imidazolines

2. Conditioning agents:

- Lanolin
- Mineral oil
- Polypeptides
- Egg derivatives
- Herbal extracts

3. Viscosity modifiers:

1. Electrolytes:

- Ammonium chloride
- Sodium chloride

2. Natural gums :

- Gum tragacanth
- Gum karaya
- Alginates

3. Cellulose derivatives

- Hydroxyethyl cellulose
- Hydroxypropyl cellulose
- Carboxymethyl cellulose

4. Carboxyvinyl polymers :

- Polymer of acrylic acid cross-linked with polyfunctional agent

5. Others:

- Ethoxylated fatty acid diesters
- Phosphate esters
- Polyvinyl pyrrolidones

4. Opacifying & Clarifying agents :

- Stearyl alcohol
- Cetyl alcohol
- Propylene glycol
- Phosphates
- Finely dispersed zinc oxide
- Milky emulsions of vinyl polymers

5. Preservatives: (Formaldehyde, Esters of para hydroxy benzoic acid)

6. Perfumes

Formula: -

Coconut oil	- 14% v/v
Olive oil	- 3% v/v
Castor oil	- 3% v/v
Potassium hydroxide	- 8% v/v
Glycerol	- 8% v/v
Sodium hexamethyl cellulose	- 0.8% v/v
Perfume	- 0.3% v/v
Sodium carboxymethyl cellulose	- 2.5% v/v
Water	- 53% v/v

Procedure for preparation of shampoo: -

- Measure the quantity of coconut oil, olive oil, castor oil & take it in a round bottom flask.
- Potassium hydroxide dissolved in 3/4th quantity of water was added to it; heat it in a water bath for 70-80°C with constant shaking for 1hr.
- Take remaining quantity of water and add sodium hexamethyl cellulose, glycerine, perfume & other ingredients.
- Then mix both the solution to form a uniform shampoo.

Types of shampoo:

1. Powder shampoo:

Formula: -

- | | |
|-----------------------|-------|
| ➤ Henna powder | 5% |
| ➤ Soap powder | 50% |
| ➤ Sodium carbonate | 22.5% |
| ➤ Potassium carbonate | 7.5% |
| ➤ Borax | 15% |
| ➤ Perfume | Q.S. |

2. Lotion shampoo:

Formula: -

- Sodium lauryl sulphate 35%
- Glyceryl monostearate 2%
- Magnesium monostearate 1%
- Water Upto 100ml
- Colour, Perfume & Preservatives Q.S.

3. Liquid shampoo:

Formula: -

Sodium lauryl sulphate	-	40%
Sodium chloride	-	2-4%
Water	-	upto 100%
Perfume, Colour & Preservatives	-	Q.S

4. Anti-Dandruff Shampoo:

Formula: -

Selenium sulphide	-	2.5%
Bentonite	-	5%
Sodium lauryl sulphate	-	35%
Water	-	Upto 100%
Colour , Perfume , Preservatives	-	Q.S

5. Conditioning shampoo:

Formula: -

- Stearyl dimethyl benzyl ammonium chloride 5.5%
- Ethylene glycol monostearate 2%
- Cetyl alcohol 2.5%
- Water Upto 100%
- Colour, Perfume & Preservatives Q.S.

Evaluation of Shampoo

1. Determination of percent of solid contents:

- A clean dry evaporating dish was weighed and added 4g of shampoo to it.
- Evaporating dish with the shampoo was weighed.
- The exact weight of shampoo was calculated & the evaporating dish with the shampoo was placed on a hot plate until the liquid portion evaporates.
- The weight of shampoo only (solids) after drying was calculated.

2. Determination of pH:

- The pH of 10% shampoo solution in distilled water was determined at room temperature using a digital pH meter.

3. Determination of viscosity:

- It is determined using Brookfield viscometer .
- 100ml of shampoo is taken in a beaker & spindle is dipped in it for about 5 minutes and then reading is taken.

4. Foaming ability & Foam stability:

- 50ml of 1% shampoo solution was put into a 250ml graduated cylinder and covered the cylinder with hand and shaken for 10 times.

- The total volume of the foam contents after 1 minute shaking was recorded.
- The foam volume was calculated after shaking the volume of the foam at 1 minute interval for 4 minutes were recorded.

5. Skin irritation test:

- Patch test technique is used here.
- A set of 6 rabbits were used for testing each material.
- The shampoo preparations are diluted between concentrations of 8-10%.
- Patches containing the shampoo preparations were applied & then removed, the skin sites were noted for any changes on the surface of skin like edema, erythema formation.
- These products are then classified as mild irritant, moderate & severe.

6. Cleaning action:

- 5g of wool yarn were placed in grease, after that it was placed in 200ml of water containing 1g of shampoo in a flask.
- Temperature was maintained at 35°C.
- The flask was shaken for 4 minutes at the rate of 50 times a minute.
- The solution was removed and sample was taken out, dried and weighed & calculated the amount of grease removed.

7. Dirt dispersion:

- Two drops of shampoo were added in a large test tube containing 10ml of distilled water.
- 1 drop of ink was added; the test tube was stoppered and shaken it 10 times.

- The amount of ink in the foam was estimated as none, light, moderate or heavy.

8. Surface tension measurement:

- Measurements were carried out with a 10% shampoo solution in distilled water at room temperature.
- Thoroughly clean the stalagmometer with purified water because surface tension is highly affected with grease or other lubricants.

Surface tension measurement is done by the following equation: -

$$R_2 = \frac{(W_3 - W_1) n_1}{(W_2 - W_1) n_2} * R_1$$

9. Eye irritation test:

- About 1% shampoo solution was dripped into the eyes of 6 albino rats with their eyes held open with the clips at the lid.
- The progressive damage to the rabbit's eyes was recorded at specific intervals over an average period of 4 seconds.
- Reaction to the irritants can include swelling of the eyelid, inflammation of the iris, ulceration, hemorrhaging and blindness.

5. Toothpaste

Toothpaste is a paste or gel dentifrice used with a toothbrush as an accessory to clean & maintain the aesthetics and health of the teeth.

Ingredients of Toothpaste

1. Abrasives:

These are used to remove food debris and residual stains of the teeth & to polish the teeth surface.

Example:

- Precipitated calcium carbonate
- Dibasic calcium phosphate
- Tricalcium phosphate
- Calcium pyrophosphate
- Insoluble sodium metaphosphate
- Anhydrous dibasic calcium phosphate
- Hydrated alumina

2. Detergents & other foaming materials:

- These are used in toothpaste for their cleansing action
- These materials lower surface tension, thereby promoting penetration of paste & helps in removal of deposits & debris.

Example:

- Sodium lauryl sulphate
- Magnesium lauryl sulphate
- Sodium lauryl sarcosinate
- Sodium lauryl sulphoacetate
- Dioctyl -sodium sulphosuccinate
- Monoglycerides

- Sulphates
- Sulphonates

3. Humectants:

- Humectants in toothpastes are used to prevent their drying out
- They also impart some plasticity to toothpastes.

Example:

- Glycerin
- Propylene glycol
- Sorbitol solution

4. Binding agents:

- Solid & liquid phases of toothpaste tends to separate in the absence of binding agents particularly during storage.
- Binding agents disperse, swell or absorb water to form viscous liquid phases.

Example:

- Natural gums (Gum Arabic, Gum tragacanth)
- Cellulose derivatives (Carboxy methylcellulose, Sodium carboxy methyl cellulose, Hydroxyethyl cellulose)
- Starch ethers
- Synthetic resins

5. Sweetening & flavoring agents :

- These are the most important ingredients in toothpaste as they have great influence on the acceptance of the product.

Example:

- Saccharin sodium (0.05-0.3%)

- Chloroform
- Peppermint
- Spearmint
- Coriander
- Caraway
- Anise
- Eucalyptus

6. Minor ingredients:

- Titanium dioxide
- pH regulators
- Cellulose ethers
- Carragheen

Formula: -



Calcium carbonate	-	60g
Glycerin	-	20g
Tragacanth powder	-	2g
Sodium lauryl sulphate	-	2g
Saccharine	-	0.5g
Peppermint oil	-	0.5ml
Methyl paraben	-	1g
Propyl paraben	-	1g
Water	-	15ml

Procedure for preparation of toothpaste :-

- Take half the quantity of water, add tragacanth powder & heat it in a water bath to get a gel.
- To the remaining quantity of water, add glycerine, sodium lauryl sulphate, preservative & mix it thoroughly to get a clear solution.

- Weigh the required quantity of saccharin and calcium carbonate solution & mix it with the help of mortar and pestle.
- To this powder, add gum tragacanth & mix well.
- Add glycerine, preservative & sodium lauryl mixture to it and triturate uniformly to get a paste.
- Finally add flavoring agent & triturate well.
- Transfer to a narrow mouthed plastic tube, seal & label.

Evaluation of Toothpaste:

1. Determination of hard & sharp edged abrasive particles :

- The paste was extruded about 15 to 20cm length from collapsible tube of each sample on a butter paper.
- Then all the samples were tested by pressing it along it's entire length by a finger for the presence of hard & sharp edged abrasive particles.

2. Determination of spreadability:

- About 1g of each sample was weighed and placed at the centre of glass plate & another glass plate was placed over it carefully.
- Above the glass plates, 2kg weight was placed at the centre of plate, avoid sliding of plate.
- The diameter of the paste in cm was measured after 30 minutes.
- The experiment was repeated 3 times & the averages were reported for all the samples.

3. Determination of fineness:

- A sample of 10g was accurately weighed and placed in a 100ml beaker.
- To this, 50ml of water was added & allowed to stand for 30minutes with occasional stirring until the toothpaste was completely dispersed.

- This solution was passed through 150micron standard sieve & sieve was washed with water.
- After washing, the residue remained on sieve was collected & dried in an oven at 105°C.
- After drying, the sample was collected carefully & weighed
- Fineness was calculated by using the following formula;

$$\text{Percentage by mass} = \frac{M_1}{M_2} * 100$$

4. Determination of pH:

- Take 10g of toothpaste in 150ml beaker.
- Add 10ml freshly boiled & cooled water and stir well to make a thorough suspension.
- Determine the pH of suspension using pH meter.

5. Determination of foaming power:

- To 5g of sample, 10ml of water was added & covered with a watch glass, allowed to stand for 30 min to disperse toothpaste in water
- The contents of the beaker was stirred & slurry was transferred to a 250ml graduated measuring cylinder, ensure that no foam was produced & no lump went into the cylinder.
- The content of the cylinder was adjusted to 50ml by adding sufficient water & temperature was maintained at 30°C
- Then the cylinder was stoppered & shaken it 12 times and the cylinder was allowed to stand for 5 min and the volume of foam was noted for all samples.

$$\text{Foaming power} = V1 - V2$$

6. Moisture content:

- 10g of toothpaste was weighed & dried it in an oven at 105°C, and then it was cooled.
- The loss of weight is recorded as percentage moisture content and calculated by given formula;

$$\% \text{ moisture} = \frac{\text{Original sample wt} - \text{Dry sample wt}}{\text{Original sample wt}} \times 100$$

7. Determination of heavy metals:

- Accurately weigh 2g of the sample in kjeldahl flask
- An acid mixture of HNO₃: HClO₄ (4:1) was added in the flask & heated continuously till the solution become colourless.
- The sample was then transferred to a 25ml volumetric flask & volume was made up with distilled water.
- A reagent blank was prepared according to the above procedure.
- The standard of lead was prepared as per the protocol & the sample was visually analysed & compared with the standard solution of lead.

6. Soaps & Syndetbars

- Soaps are sodium or potassium salts of long chain fatty acids. When triglycerides in fat/oil react with aqueous NaOH or KOH, they are converted into soap & glycerol. This is called alkaline hydrolysis of esters.
- Since this reaction leads to the formation of soap, it is called saponification process.
- The word "Syndet" is derived from 'synthetic' combined with 'detergent'. Technically, it refers to the binding that occurs between different detergents, also called surfactants.
- Syndet soap surfactants are derived from oils, fats or petroleum products that are processed from a wide range of chemical processes other than traditional saponification.

Ingredients for making soap:

1. Fats & Oils:

A Fat mixture containing saturated & unsaturated, long & short chain fatty acids in proportion are used in manufacture of soap

Example:

- A very common mixture for manufacture of soap is 75% tallow & 25% coconut oil.
- Saturated fatty acids with 12 to 18 carbon atoms include lauric, myristic, palmitic, stearic & oleic acids are used.

2. Alkali:

An important raw material in soap making is caustic soda.

Example:

- Soda ash can saponify fatty acids & used as a builder in laundry soaps.
- Caustic potash (KOH) is also used in making soft soaps.
- Potassium carbonate can be also used to saponify fatty acids

3. Other additives:

- Water
- Salt (Sodium chloride)
- Anti-oxidants
- Whiteners
- Perfumes
- Pigments & Colours

Ingredients for making syndetbars

- Sodium cocoyl isethionate
- Sulphosuccinates
- Alpha olefin sulphonates
- Alkyl glyceryl ether sulphonate
- Sodium cocoyl monoglyceride sulphate
- Betaines

Manufacture of soaps & Syndetbars - Soap is manufactured by 3 types of processes:

1. Semi-boiled & cold process :-

- In these processes, fat is reacted with strong alkali. After reaction is complete, the entire mass is saponified without the separation of free glycerine.
- Here, the fat charge is simply heated with required amount of caustic soda.
- After saponification, other ingredients like soap builders are added.
- If the batch is made in a kettle, it may be actually boiled but if the batch is made in a crutcher, the temperature must be kept little below the boiling point.

2. Full-boiled process: -

Here, large cylindrical kettles with cone bottoms equipped with open & sometimes with closed coils for steam are used.

It includes:

Saponification reaction: This is carried out by boiling fat with aqueous alkali together with open steam.

Graining out & washing: -

- After saponification, the soap is grained out by the addition of salt to the boiling mass.
- Dry salt as well as brine may be used.
- The soap rises to the top of kettle in form of rough mass as soap which is called 'kettle wax'
- The washings are carried out by adding water to the kettle mass.

Finishing or Fitting operation:-

- The final operation in soap boiling is fitting operation.
- Here, the soap is boiled with water & upon standing, the batch will separate into upper layers consisting of neat soap and lower layer consisting of niger which retains most of the colouring materials, metallic salts & other impurities.

3. Continuous saponification

- Here, fats are first converted into fatty acids & glycerine using high pressure at 500°C in continuous fat splitting process and fatty acids are reacted with alkali to form soap.
- This method is suitable for large scale production.

Formula for soap formulation:

Soap base	-	99.3-98.6%
Titanium dioxide	-	0.2-0.4%
Perfume	-	0.5-1%
Colour	-	Q.S

Formula for Syndetbars formulation: -

Sodium glyceryl monolaurate sulphate	-	78%
Glyceryl monostearate	-	10%
Cetyl alcohol	-	9%
Lanolin derivative	-	1%
Corn starch	-	2%
Perfume, colour, Anti-oxidant	-	Q.S

Evaluation of Soaps & Syndetbars

1. Determination of clarity, color & odor: - Clarity & colour was checked by naked eyes against white background, odour was smelled.

2. Determination of pH: -

- The formulation were dissolved in 100ml of distilled water & stored for 2hrs.
- The measurement of pH of the formulation was done in previously calibrated pH meter.

3. Determination of foam height:-

- 0.5g of sample of soap was taken dispersed in 25ml distilled water.
- Then transferred it into 100ml measuring cylinder. Volume was adjusted to 50ml with water.
- Shaken the above sample solution 25 times and measured the foam height.

4. Determination of percentage of free alkali

- About 5g of sample was taken in a conical flask and added it into 50ml of neutralized alcohol.
- It was boiled under reflux on a water bath for 30min, cooled & 1ml of phenolphthalein solution was added.
- It was then titrated immediately with 0.1N Hcl.

5. Determination of alcohol insoluble matter:-

- 5g of sample was taken in a conical flask, added it to 50ml of warm ethanol & shaken vigorously to dissolve.
- The solution was filtered through a filter paper with 20ml warm ethanol and dried it at 105°C for 1 hr.

PERFUMES

Introduction

- The word perfume is derived from the Latin word "per fumum", meaning through smoke.
- Perfume is a mixture of fragrant essential oils or aromatic compounds, fixatives and solvents used to give the human body, animal, food objects and living spaces a pleasant scent.
- Perfumes are supposed to release a continuous pleasant fragrance that will provide a long lasting feeling of freshness.
- Initially it was only used for religious purpose but now it has become an ornament of sophistication for both men and women.

Notes in Perfume

Perfume is described in a musical metaphor as having three sets of notes, making the harmonious scent accord.

These notes are created carefully with knowledge of the evaporation process of the perfume.

The three notes are:

- a. Top notes
- b. Middle notes
- c. Base notes

a. Top notes - They are generally the lightest of all notes and recognized immediately after application. Top notes consist of small, light molecules with high volatility that evaporate quickly. Common fragrances of top notes include citrus (lemon, orange zest), light fruits (grape, berries), and herbs (clary sage, lavender).

b. Middle Notes - The middle notes or the heart notes. Makes an appearance once the top notes evaporate. The middle note compounds form "heart" or main body of a perfume and act to mask the often unpleasant initial impression of base notes, which become more pleasant with time. Common fragrances of middle notes include rose, lemon, nutmeg, jasmine etc.

c. Base notes: Base notes or bottom or dry notes appear while middle notes are fading. The base and middle notes together are the main theme of a perfume. Base notes bring depth and solidity to a perfume. Common fragrances of base notes include sandalwood, vanilla. Amber and musk.

Classification of perfumes - Perfumes are classified into five major groups on the basis of concentration of fragrance and duration of lasting:

Class	% of aromatic compound	Duration (hours)
Parfume(perfume)	20-30	6-8
Eau de parfume	15-20	4-5
Eau de toilette	5-15	2-3
Eau de cologne	2-4	2
Eau fraiche	1-3	2

Perfumes can further be classified into following classes:

- **Bright Floral:** Fragrance from one or several flowers. E.g. Estee lauder's Beautiful
- **Green:** Fragrance from cut grass or leaf. E.g. Calvin Klein's Eternity
- **Aquatic:** A clean smell reminiscent of ocean. E.g. Davidoff Cool Water
- **Citrus:** Has freshening effect. E.g. Faberge Brut
- **Fruity:** Aromas of fruits other than citrus e.g. Ginestet Botrytis

- **Gourmand:** Scent with edible or desert like qualities. E.g. Thierry Mugler's Angel.

Composition of Perfumes - Perfumes are mainly composed of:

- a. **Essential oils:** Derived from natural aromatic plant extracts and/or synthetic aromatic chemicals. E.g. limonene, linalool, geraniol, citral etc.
- b. **Fixatives:** Natural or synthetic substances used to reduce the evaporation rate. E.g. benzyl benzoate, benzyl alcohol etc.
- c. **Solvents:** The liquid in which the perfume oil is dissolved in is usually 98% ethanol and 2% water. Alcohol allows fragrance to spread along with it and does not permit microbial growth in the perfume.

Manufacturing process - Perfumes can be manufactured by following steps:

1. Collection
2. Extraction
3. Blending
4. Aging

1. Collection - Before manufacturing process begins the sources of suitable fragrances are collected in the manufacturing centre.

2. Extraction: Oils are extracted from plants and other substances by several methods like:

- a. **Steam distillation:** steam is passed through plant materials held in a still, whereby the essential oil turns to gas. This gas is then passed through tubes, cooled, liquefied and collected.
- b. **Solvent extraction:** The flower parts are dissolved in benzene or petrolatum that retains the fragrance of the flower. Alcohol is used

to dissolve the fragrance and heated to obtain it after evaporation of alcohol.

- c. **Enfleurage:** Flowers are kept in glass sheet with grease that absorb the fragrance of flowers and then extracted with alcohol.
- d. **Expression:** The citrus fruits or plants are manually or mechanically pressed until all the oil is squeezed out.

3. Blending:

- Once the perfume oils are collected, they are ready to be blended together according to a formula determined by a master in the field, known as a "nose."
- After the scent has been created, it is mixed with alcohol. Most perfumes are made of about 10-20% perfume oils dissolved in alcohol and traces of water.

4. Aging: Fine perfume is often aged for several months or even years after blending to ensure that the correct scent has been achieved.

Natural Perfumes

- Ambrette seed oil
- Coco absolute
- Coffee absolute
- Pink lotus absolute

Chemical Odor

- Allyl hexanoate: Pineapple
- Benzaldehyde: Bitter almond
- Cinnamic aldehyde: Cinnamon
- Ethyl propionate: Fruity
- Limonene: Orange
- Methyl salicylate: Wintergreen

Aroma Compounds

Aroma compounds are classified on basis of functional groups:

- Alcohols
- Aldehydes
- Amines
- Esters
- Ethers
- Ketones
- Lactones
- Terpenes
- Thiols

Perfumes ingredients listed as allergens in EU regulation

The Cosmetics Regulation 1223/2009 lists the 26 most known allergic substances. These substances must appear on the label of your cosmetic product when present in the finished formula.

Allergen - An allergen is commonly defined as "any substance, often a protein that induces an allergy."

Category	No. of individual fragrances	No. of natural extracts	Examples
Established in humans	54	28	Citral, Geraniol, Turpentine oil,
Established in animals	18	1	Isocyclocitral, Dibenzyl ether, Jasminum Sambac flower extract
Likely	26	-	Ethyl vanillin, Methyl cinnamate, Phytol
Possible	35	13	Benzyl acetate, Phenylpropanol, Illicium verum fruit oil

Ingredients	Use	Side effects
Sandal wood	Fragrance	Hypersensitivity
Limonene	Slightly astringent smell	Irritates the skin
Benzyl alcohol	Fixative	Skin irritant causing redness and pain
Benzyl benzoate	Fixative; sweet balsamic odor	Skin irritation like blister, itching, scaling, redness.
Acetone	Solvent	Inhalation cause dryness of mouth & throat
Ethyl acetate	Solvent	Defatting effect on skin & may cause drying & cracking

Cosmetics directive required that the presence of any of these 26 substances be indicated in the list of ingredients when its concentration exceeds 0.001% in leave on products and 0.01% in rinse off products.

Such labeling allows patients who are allergic to one or more of these 26 fragrance chemicals to avoid products containing them.

List A: Fragrance chemicals, which according to existing knowledge, are most frequently reported and well-recognised consumer allergens.

- Amyl cinnamal
- Amylcinnamyl alcohol
- Benzyl alcohol
- Benzyl salicylate
- Cinnamyl alcohol
- Cinnamal
- Citral
- Coumarin
- Eugenol
- Geraniol
- Hydroxycitronellal
- Hydroxymethylpentyl-cyclohexenecarboxaldehyde
- Isoeugenol

List B: Fragrance chemicals, which are less frequently reported and thus less documented as consumer allergens.

- Anisyl alcohol
- Benzyl benzoate
- Benzyl cinnamate
- Citronellol
- Farnesol

- Hexyl cinnamaldehyde
- Lilial
- d-Limonene
- Linalool
- Methyl heptine carbonate
- 3-Methyl-4-(2,6,6-trimethyl-2-cyclohexen-1-yl)-3-buten-2-one

Controversial Ingredients of Cosmetics

Parabens

- Parabens are widely used preservatives and it is used in various pharmaceuticals products.
- They are a series of P- hydroxybenzoates or esters of P- hydroxybenzoic acid.
- It is used in many cosmetic formulations such as shampoos, shaving gels, commercial moisturizers, topical/ parenteral preparations, tanning solution, toothpaste and also in a food additives.
- Its antibacterial mode of action is not well understood but it is more effective against broad spectrum of microorganisms by inhibiting the synthesis of DNA and RNA.
- Common parabens include are: methyl paraben, propyl paraben, ethyl paraben and butyl paraben.
- Parabens are controversial due to following factors given by American cancer society and FDA:

According to research, it is found that it is relatively safe, posing only a negligible risk to the endocrine systems.

- Allergic reactions
- Breast cancer
- Estrogenic activity
- Sun exposure
- Parabens can mimic hormones in the body and disrupt functions of the endocrine system.
- Paraben can enter in your body through your skin when you applied it on the skin.
- According to Mercola, the body can absorb as much as 5 pounds of cosmetic chemicals every year.

Health concerns:

Breast cancer

- Cornell university reports that a high lifelong exposure to estrogen can increase risk of it.
- Paraben accumulate in fat cells including breast tissues and leads to cause breast cancer.
Eg.: methyl paraben

Early puberty

- Paraben are having ability to adversely affect the endocrine system.
- It also results in early puberty in adolescent girls and boys.
- Paraben causes defectiveness in male reproductive system, results in decrease in sperm level concentration.

Regulations:

- The European Scientific Committee on Consumer Safety (SCCS) reiterated in 2013 that methyl paraben and ethyl paraben are safe at the maximum authorized concentration [upto 0.4% for one ester or 0.8% used in combination]
- They concluded that the use of butyl paraben and propyl paraben as preservatives in finished cosmetics products is safe to the consumer.

Alternatives for Parabens

- Phenoxyethanol- used in the EU at upto 1.0% in all product categories.
- Sodium benzoate in combination with potassium sorbate, upto 2.5% in rinse off products, 1.7% in oral care products, 0.5% in leave-on products.

Formaldehyde Liberators

- Formaldehyde is a colorless, strong smelling gas used in a wide range of industries and products including building materials, cabinets furniture, walls and personal care products.
- Formaldehyde and formaldehyde releasing preservatives are found in many personal care products, particularly in shampoos, liquid baby soaps, nail polish, nail glue, hair gel and etc.

Health concerns:

- Cancer
- Skin irritation

Vulnerable populations:

- Infants
- Hair salon workers
- Nail salon workers

Cancer

- Formaldehyde is considered as a known human carcinogenic by many experts and government bodies.
- A 2014 study found that formaldehyde initiates and promotes tumour formation.
- It enters into a body through an inhalation of formaldehyde containing formulations leads to cause cancer.

Irritation

- Formaldehyde, in the 2015 by American contact dermatitis society considered as contact allergen of the year.
- It also results in occurrence of sensitivity.

Regulations

- Banned from use in cosmetics and toiletries in Japan and Sweden.
- EU allows the use of quaternium 15 upto 0.2% as preservatives in cosmetics products.

How to avoid:

- Skin hair smoothing products especially those sold in salons, as salon based products are exempt from labelling laws.
- Don't use expired cosmetics products or store cosmetic products in the sun because this can cause more formaldehyde to be released.
- Choose nail products that are labeled formaldehyde free or toxic free (formaldehyde, toluene and DBP).

What to look on the label: Quaternium-15

- a. It is a quaternary ammonium salt used as a surfactant and preservatives in many cosmetics and pharmaceutical products.
- b. It is the most sensitizing of FRP and is found in mascara, lotion and shampoo.

Glyoxal

- a. Glyoxal is the smallest dialdehyde and organic compound which is yellow colored liquid that evaporates to give a green colored gas.
- b. It is found in conditioner, lotion, nail polish, etc
- c. CIR expert panel has declared that glyoxal is an skin allergen.

Bromopol

- a. It is found in nail polish, makeup remover, moisturizer and bodywash.
- b. It is safe in concentration less than 0.1% but cannot be found in formulations with the FRP amine.
- c. Mixing bromopol and amines produce nitrosamines which have been found to penetrate to skin and cause cancer.

1,4- dioxane

1,4-dioxane is generated through a process called as 'Ethoxylation', in which ethylene oxide a known breast carcinogens is added to other chemicals to make them harsh.

Example:

- Sodium laurel sulphate - a chemical that is harsh on the skin, is often converted to the less harsh chemical sodium laureth sulphate.
- The conversion process can lead to contamination of this ingredients with 1,4-dioxane.

Other common ingredients that may be contaminated by 1,4- dioxane include PEG compounds and chemicals that include the clauses 'Xynol', 'cetearth' and `Oleth'.

Most commonly 1,4-dioxane is found in products that create suds like shampoo, liquid soap and bubble bath.

Environmental Working Group's analysis suggests that 97% of hair relaxers, 57% of baby soaps and 22% of all products in skip deep may be contaminated with 1,4-dioxane.

According to survey of campaign for safe cosmetics in 2017, showed that popular brands of childrens bubble bath and body wash contained 1,4- dioxane.

Health concerns:

Cancer-

- Research shows that 1,4-dioxane readily penetrate the skin.
- It is considered as a probable human carcinogen by the U.S.Environment Protection Agency.
- It is listed as an animal carcinogen by the National Toxicology Program.
- It is included on California's proposition 65 list of chemicals known or suspected to cause cancer or birth defects.
- It is also leads to cause organ toxicity.

Vulnerable population:

- Infants
- Teenagers
- Pregnant womens

Regulations: As per regulations, it is found that limit for dioxane safe for human is 10 ppm.

How to avoid:

- There are several ways to avoid the formation of 1,4-dioxane in the ingredients:
- The FDA does not require 1,4-dioxane to be listed as an ingredients on product labels because the chemical is a contaminant produced during manufacturing.
- Without labeling, there is no other options to know for certain whether a product contains 1,4-dioxane, making it difficult for consumers to avoid it.
- Alternative processes to ethoxylation do exists but many companies don't take advantage of them.

- Vacuum stripping can remove 1,4- dioxane from an ethoxylated product.
- Organic standard do not allow ethoxylation and some conventional companies such as Johnson and Johnson have agreed to alter the process the results in this contamination.
- A study by the Organic Consumer Association shows that 1,4- dioxane is non- existent in a variety of cosmetics certified under the USFDA National Organic program.
- Hence, a good way to avoid exposure to this chemical is to buy products that have been certified under this program.

Unit 4 - Design of Cosmeceuticals products

Sun Protection

- Exposure to sunlight can have both beneficial and harmful effects on the human body, depending on the length and the frequency of exposure, the intensity of sunlight and the sensitivity of individual concerned.
- The most effect of exposure to the rays of sun is erythema of skin, followed by the formation of a tan which seems to be a problem of civilized world.
- Such damage following repeated, lifetime exposure to solar UV lead to skin cancers and a myriad of degenerative events responsible for the visible signs of skin aging.
- Recent years have seen a very rapid increase in knowledge concerning the etiology and prevention of solar damage. Since exposure to UV radiation in sunlight is associated with deleterious dermatological events, it is logical that reducing solar UV exposure will diminish such damage to the skin.
- The complete avoidance of solar UV is neither achievable nor entirely healthy. ie:, it is known that exposure to sunlight has health benefits including production of vitamin D.
- A "safe sun" strategy has been developed and promoted by healthcare professionals worldwide. An important part of this "safe sun" strategy is the use of sunscreens.
- Once the energy from UV is absorbed in the skin, it may produce new chemical entities, e.g., 60,40-DNA photoproducts, free radicals, etc., or dissipate the excess energy as heat or phosphorescence.
- This absorption and subsequent conversion of energy contribute to the processes involved in the etiology of skin cancer and photo aging.

UV Index

Model that relates ozone levels to UV radiation on the ground, forecasted cloud amounts, and the elevation of the forecast cities

3 types given off by the sun:

1. UVC

- a. 100-290 nm.
- b. Absorbed by the ozone and doesn't affect the skin.

2. UVB

- a. 290-320 nm.
- b. Responsible for most severe damage.
- c. Affects the outer layer of the skin and is the primary cause of sunburn.
- d. Most intense between 10-2 and during the summer months.
- e. Direct impact on cell DNA and proteins.

3. UVA

- a. 320-400 nm.
- b. Penetrates deeper than UVB.
- c. Affect connective tissues by producing reactive oxygen species: produce immune suppression
- d. Responsible for tanning, photo aging photo carcinogenesis

Damaging Effect

1. Collagen Breakdown

- a. UV radiation increases breakdown and causes abnormal elastin accumulation.
- b. Accumulation creates solar scars and wrinkles develop.

2. **Free Radicals** - Creates free radicals which damages cell function and alters genetic material.
3. **Immune System** - Certain chemicals are released which suppresses the system

Adverse effects of sunlight

Solar radiation can have both short term and long term effects.

A. Sun burn-short term - It is a temporary damage to the epidermis.

The four degrees of sun burns are:

a. Minimal preceptive erythema

- Slight, but discernable red or pink coloration to the skin.
- Produced in 20 minutes.

b. Vivid erythema

- A bright red colour of the skin, no pain.
- Produced in 50 minutes.

c. Painful burn

- Characterized by vivid erythema and pain
- Produced in 100min

d. Blistering pain

- High pain accompanied by vivid erythema with blistering and peeling.
- Produced in 200min.

B. Chronic Exposure

1. Premature aging
2. Transient dermatitis to skin cancer

Sun Screen

Definition - Sunscreens are topical applied products that absorb or reflect some of the sun's ultra violet radiation and thus helps protect against sun burn:

- Are agents which absorbs 95% or more of UV radiation within wavelength 290-320nm.(UV B)
- Sunscreens offer full protection against UVB, but not UVA.
- Very few sunscreen ingredients protect against UVA.

Ideal characters of sunscreens

In order to ensure optimal patient compliance, an ideal sunscreen would be:

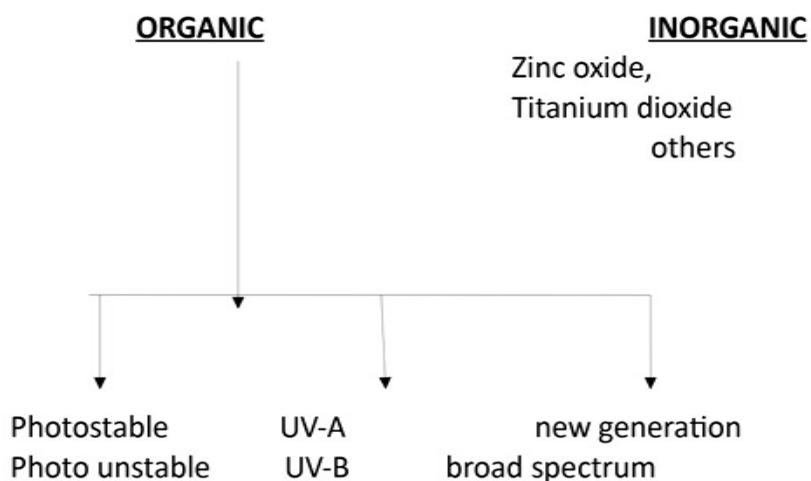
- A combination of physical and chemical agents
- Broad spectrum
- Cosmetically elegant
- Substantive
- Non irritant
- Hypoallergenic
- Economical

Functions

- Prevents solar UV from interacting with skin chromophores is the primary function of sunscreens.
- Sunscreen products are quite simple; they absorb/reflect/scatter UV radiation from sunlight before this energy can be absorbed by chromophores residing in the skin.
- Should have health benefits.

Classification of sunscreens

The last FDA sun screen monograph was issued in 1999, according to this monograph sunscreens are classified as organic and inorganic:



A. Organic/chemical agents

- Organic uv filters are active ingredients that absorb uv radiation with in a particular range of wavelength, depending on their chemical structure.
- Once the uv filter absorb energy,it moves from a low energy ground state to a high energy excited state.
- From this excited state, any of the following 3 process may occur ,depending on the ability of the filter to process the energy it has absorbed:
 1. **Photo stable filter** - Filter dissipates its absorbed energy to the environment as heat energy, and return to the ground state. It is fully capable of absorbing UV energy again.
 2. **Photo unstable filter** - Filter undergoes change in its chemical structure, or it is degraded after absorbing UV energy .it is not capable of absorbing UV energy again

3. Photo reactive filter - In its excited state the filter interacts with surrounding molecules, including other ingredients of sunscreen, oxygen, and skin proteins, and lipids.

Organic sunscreens further classified into:

1. UV B filter

- PABA derivative-padimate o
- Cinnamate-octinoxate, cinoxate
- Salicylate-octisalate, homosalate
- Octocrylene

2. UV A filters

- Benzophenones-oxybenzone, dioxybenzone
- Avobenzone, Meradimate
- ✓ Avobenzone is the only sunscreen agent that has absorption peak in UVA1 spectrum, making it very effective UVA filter, but it has disadvantage of photounstable nature.
- ✓ This can be overcome by adding agents that photostabilise avobenzone.

3. Newer generation broad spectrum (UV A+UVB) filters

- Ecamsule (Meroxy SX)
- Bemotrizinol (Tinosorb S)
- Bisotrizol (Tinosorb M)

B. Inorganic sunscreens

1. Zinc oxide
2. Titanium dioxide
3. Others-iron oxide, kaolin, calamine, talc

Sun protection factor (SPF)

- Sunscreens have been tested and rated for their SPF value.
- SPF is mainly the measure of the sunscreen product's ability to shield against UVB rays.
- SPF is the ratio of the UV radiation does it takes to cause a barely detectable sunburn on a person treated with sun screen product(2mg/cm²) compared to that required for untreated skin
Eg: If it takes 10 min to burn without a sunscreen and 150 min to burn with a sun screen.
SPF of that sunscreen = $150/10 = 15$
- Specifically, the minimum erythema dose (MED) is determined for each panelist in an SPF test. This is the time/dose of solar-simulated UV needed to produce a uniform, barely perceptible redness in the skin.

The following table compares how the skin receives a sun burning without sunscreen protection, and with protection from sunscreen with different SPF values:

SPF values	Sun burning dose over time (%)		
	10 min	30 min	150 min
1(noprotection)	100%	-	-
15	6.6%	20%	100%
30	3.3%	10%	50%
50	2%	6%	30%

To achieve the SPF and breadth of UV protection, a combination of UV filters is selected based on their individual absorption profiles and other physiochemical characteristics.

Table 2 Approved UV Filters

UV filter	Up to % concentration
Aminobenzoic acid or PABA	15
Avobenzene or butyl methoxydibenzoylmethane	3
Cinoxate	3
Dioxybenzone	3
Homosalate	15
Menthyl anthranilate or meradimate	5
Octocrylene	10
Octyl methoxycinnamate or octinoxate	7.5
Octyl salicylate or octisalate	5
Oxybenzone or benzophenone-3	6
Octyl dimethyl PABA or padimate O	8
Phenylbenzimidazole sulfonic acid or ensulizole	4
Sulisobenzene	10
Titanium dioxide or TiO ₂	25
Trolamine salicylate	12
Zinc oxide or ZnO	25

From a formulation perspective, the current list of UV filters can be categorized into one of four groups based on the physical properties of the active:

- Polar oils, e.g., octinoxate, octisalate, homosalate, and octocrylene
- Oil soluble crystalline solids, e.g., benzophenones
- Water soluble salts, e.g., ensulizole
- Insoluble powders/particulates, e.g., zinc oxide and titanium dioxide
- Polar oils tend to make the product feel greasy and oily, especially at high concentrations.
- Oil soluble crystalline solids require high levels of oily solvents/emollients to dissolve them and keep them from

crystallizing in the product over time, and hence make the product feel greasy and oily.

- Water soluble salts tend to reduce the capability of most aqueous polymeric thickeners. This, in turn, leads to the use of much higher polymer levels to achieve a target product thickness, and these high polymer levels make the product feel sticky and heavy on the skin.
- Insoluble powders/particulates can make the product feel dry and draggy, and often can lead to an undesirable white appearance on the skin.

Selection of sunscreen

- A fair skin that burns easily, should select a high -SPF broad spectrum sunscreen labeled as SPF50+
- A skin that tans readily, could choose a sunscreen with an intermediate SPF of 15+
- A darkly pigmented skin and do not and do not suffer from photosensitivity or abnormal pigmentation, no need of pigmentation

New sunscreen technologies

1. Sun spheres

- Sun spheres are styrene/acrylate copolymers that do not absorb UV irradiation but enhance the effectiveness of the active sunscreen ingredients.
- The sun sphere polymer beads are filled with water, which migrates out of the particle, leaving behind tiny air filled spheres.
- This will have low refractive index than the dried sunscreen film.
- As a result scattering of UV radiation occurs, increase the probability of contact to the active UV filter in the sunscreen.

2. Micro Encapsulation

- Active sunscreen ingredients are entrapped within a silica shell, as a result of which allergic or irritant reactions to the active ingredients can be minimized
- Incompatible sunscreen ingredients can be safely combined, without loss of efficiency

3. Regulatory aspects

Each authority has developed specific lists of allowed UV filters and their own systems to calculate test and label the SPF factor.

European Union, Switzerland, Norway and Iceland

- In Europe, sunscreens are considered as cosmetics, the system is constantly updated as technical progress evolves, on the basis of the Scientific Committee on Consumer Safety's (SCCS) recommendations on safety of ingredients.
- The Committee provides recommendations on health and safety risks of non-food consumer products (e.g. cosmetic products and their ingredients) and services
- These recommendations usually lead to an update of regulations, whose annexes have to be considered for formulating a sunscreen.

These are:

- a. Cosmetic product safety report
- b. List of substances banned in cosmetic products
- c. List of substances which cosmetic products must not contain according to restrictions
- d. List of colourants allowed in cosmetic products
- e. List of preservatives allowed in cosmetic products
- f. List of UV filters allowed in cosmetic products

- This recommendation sets some specific guidelines regarding the UVA/UVB ratio of protection,
- Minimum efficacy - lowest allowed claim SPF 6.
- UVA protection must be at least 1/3 of UVB protection (UVA/UVB ratio).
- Critical wavelength for testing is at least 370 nm.
- Precautions and usage instructions - recommendations for protection must be shown in labeling.
- Testing made according to SPF + UVA method (in vitro systems preferred for ethical reasons).

USA

- The sunscreens are classified in the USA as "over the counter" (OTC) drugs and the main reference documents are the Final Rule (5) published in 2011 and the Sunscreens Innovation Act published in 2014.
- The Sunscreen Innovation Act has been approved with the declared purpose to make some of the new filters already in use in the EU and in other countries available to the US market
- The Final Rule deals with are mostly labeling matters.
- Label wording is strictly regulated, such as
 - Where and how to write relevant information like water resistance or, in the case of sunscreens with a broad spectrum (SPF >15),
 - Which products can bear the claim: "if used as directed with other sun protection measures, (the product) decreases the risk of skin cancer and early skin ageing caused by the sun".
- One part of the regulatory system for sunscreens is represented by the monograph, which essentially provides a standard for active ingredients.

- If a monograph has been issued for a product, all a company has to do to be allowed to market it is to demonstrate that it has met the standards of that monograph.
- This method allows new formulas to be introduced to the market and is different from the aforesaid regulations on new ingredients. •
- Twenty-four Ingredients are currently regulated by the FDA through its various final monographs

Canada

- Health Canada has issued a regulatory framework for cosmetics which is a blend of the EU and US regulations.
- Cosmetics are regulated under a framework that resembles the European one,
- Sunscreens are classified in different ways according to the single ingredient and may fall either under the Natural Health Product or the Drug Products regulations.
- A monograph (last revision issued on 23/06/2015) entitled "Sunscreen monograph" collects all the points referred to this class of products. For the marketing of a product a Product Licence Application must be filed.
- Canada's monograph includes sprays and powders. It refers to all products including those referred to as secondary sunscreens.
- Specific provisions are given for labeling broad spectrum and UVA protection, whereas certain claims are not allowed, unless supported by existing scientific data.

Asean

- The cosmetic directive for ASEAN (7) countries is quite the same as the EU regulatory framework. Differences may arise due to a delay in adopting the EU approach and decisions.

- ASEAN scientific committee act like the SCCS, but the adoption of the directive is not homogeneous over the Member Countries.
- Do not stay too long in the sun while using a sunscreen.
- Re-apply frequently to maintain protection, especially after sweating, swimming or toweling.
- The use of sunscreens is one way to reduce the dangers from sun exposure.
- Instructions for use to ensure that sufficient quantity is applied, e.g. pictogram, illustration, etc.
- Over-exposure to the sun is a serious health threat.

Japan

- The classification of sunscreens as cosmetics simplifies the process for these products, as a great part of the requests are linked to the JCIA self-regulating standards.
- The UV filters are among the ingredients that are regulated.
- The minimum SPF may even be as low as just 2 while the maximum is 50+. The request to add specific warnings such as "reapply every 2-3 hours and reapply after toweling"

China

- Identified a classification as "special use cosmetics" for sunscreens.
- This means that registration of a product with the State Food and Drug Administration (SFDA) will check the safety and efficacy of the product, along with the SPF and other parameters.
- The list of permitted filters is the same as the EU, which is due to the close exchange between the EU commission and Chinese authorities looking for harmonisation of regulations.
- The maximum level is SPF 50 while for UVA the rates are PA +, ++.

- An issue which is always in discussion is China's request cosmetics for animal testing and, in this case the in-vivo UVA mandatory testing.

Australia

- The sunscreen regulations in Australia are full of important requirements that reflect the high level of attention towards sun protection in Australia.
- This specific need is due to the geographical and meteorological position of this country.

Sunscreens are classified mainly as therapeutic and cosmetic sunscreens.

Therapeutic sunscreens include:

- Primary sunscreens with SPF 4 or more
- Secondary sunscreens - except those regulated as cosmetics
- Primary or secondary sunscreens with SPF 4 or more that contain an insect repellent.
- Products containing sunscreen agents with SPF less than 4 that are currently defined as list able sunscreens molecule.
- These products are regulated as cosmetics by the National Industrial Chemicals Notification & Assessment Scheme (NICNAS) rather than therapeutic goods, which are classified by the TGA.
- Among the specific rules for Australia, a norm regulates the allowed filters and their amount, and there are a number of specific rules for claims. These are different between the two categories.
- Therapeutic sunscreens with a broad spectrum protection (SPF of 30 or higher) may have these indications on the label:
 - 'May assist in preventing some skin cancers.'
 - 'May reduce the risk of some skin cancers.'

- 'Can aid in the prevention of solar keratoses.'
- 'Can aid in the prevention of sunspots'

Dry Skin

- Dry skin is the condition when the oil glands do not supply enough lubrication to the skin. As a result the skin becomes dehydrated
- Dry skin can range from the simple, environment related itchy skin lacking some moisture to a geriatric hyperkeratosis.
- Most products developed for skin care are claimed to be emollient, moisturizing, and with a variety of useful purpose.
- Dry skin cosmetic preparations are moisturizing creams and moisturizing lotions. The difference in the two preparations virtually lies in its viscosity and hence packaging.
- In addition to environmental and external factors, biological factors like aging, hormonal disturbances etc play a significant role in skin drying.
- Aging leads to changes in the lipid composition of the cellular membrane. Phospholipids appear to decrease with increase in free cholesterol.
- Such changes suggest that the skin care products should be based in natural components or their synthetic counterparts and includes phospholipids, aminoacids, and essential hormonal factors.

Method to achieve skin moisturization - Skin moisturization can be achieved by:

- Simple occlusion whereby a barrier material is applied to the skin surface which physically prevents the loss of water from the stratum corneum.

- By the use of humectant material, water binding ingredients are applied to the integument and share water with the horny layer, thereby increasing pliability.
- Method of protection whereby the onset of dryness is prevented before it becomes a problem.
- Combination of any of these methods wherein materials are utilized which are designed to work synergistically to prevent and/or treat dry skin.

Ingredients

1. Emollients

- Emollients are ingredients that remain in the stratum corneum to act as lubricants. They help maintain the soft, smooth, and pliable appearance of the skin.
- Emollients are often thought of as "filling in the crevices" between corneocytes that are in the process of desquamation (shedding).
- The type of emollient used in a moisturizer plays a key role in its "skin slip," which is the smooth feeling imparted to the skin after application.

Acetamide MEA	Lactamide
Agarose	Lactic acid
Ammonium lactate*	Maltitol
Arginine PCA	Melibiose
Betaine	Panthenol
Butylene glycol	Pantolactone
Copper PCA	PCA
Corn glycerides	Polyglucuronic acid
Diglycereth-7 Malate	Polyglyceryl methacrylate
Diglycerin	Propylene glycerol
Dimethyl imidazolidinone	Saccharide hydrolysate
Erythritol	Sea salt
Gelatin	Sesame amino acids
Glucose	Sodium aspartate
Glucuronic acid	Sodium lactate*
Glucuronolactone	Sodium malate
Glutamic acid	Sodium PCA*
Glycereth-12	Sodium polyaspartate
Glycerin*	Sorbitol
Honey extract	TEA-lactate
Hyaluronic acid	Triglycereth-7 citrate
Hydrolyzed wheat starch	Urea
Hydroxyethyl sorbitol	Xylose

2. Occlusive's

Acetylated castor oil	Hydrogenated lanolin
Acetylated lanolin alcohol	Isocetyl myristate
Behenyl isostearate	Lanolin linoleate
Beeswax	Lauryl cocoate
C12-18 acid triglyceride	Lecithin
C20-40 alcohols	Mineral Oil
C20-40 alkyl dimethicone	Myristyl myristate
C16-36 alkyl stearate	Neatsfoot oil
C18-70 isoparaffin	Octyldodecyl stearate
C20-24 olefin	Oleyl linoleate
C10-18 triglycerides	Palm kernel wax
Candelilla	P:
Canola oil	Pentaerythrityl tetracoco
Caprylic/capric triglyceride	Petro
Carnauba	Propylene glycol diol
Cetearyl methicone	Shark liver oil
Cetyl ricinoleate	Soybean lipid
Cholesteryl oleate	Stearyl stearate
Cyclomethicone	Squalane
Decyl myristate	Tall oil
Dimethicone	Tocopherol
Distearyl ether	Trihexyldecyl citrate
Glycol dioleate	Triisostearin
Hexyldecyl isostearate	Vegetable oil
Hydrogenated castor oil	

- Occlusive increase the water content of the skin by slowing the evaporation of water from the surface of the skin.
- These ingredients are often greasy and are most effective when applied to damp skin.
- Mineral oil is often used because of its favorable texture, but it is not as effective at preventing evaporation of water as many other occlusive.
- Lanolin is expensive and potentially irritating.
- Silicone derivatives (dimethicone and cyclomethicone) are not greasy but have a limited moisturizing effect. They are often added to petroleum to make it feel less "greasy."

3. Humectants

Humectant compounds attract water "from the inside out," that is, from the dermis with passage into the upper epidermis

Acetylated lanolin
Acetyl trihexyl citrate
Avocado sterols
Butyl myristate
C 14-15 alcohols
C 12-13 alkyl ethylhexanoate
Caprylyl glycol
Castor oil
Cetyl acetate
Cetyl oleate
C 14-16 glycol palmitate
C 12-20 isoparaffin
Cyclomethicone
Decyl oleate
Diethylhexyl adipate
Diethylhexyl malate
Diisodecyl adipate
Diisopropyl dilinoleate
Dimethicone copolyol
Dipropyl adipate
Ethylhexyl palmitate
Ethyl linoleate
Glyceryl dioleate
Glyceryl ricinoleate
Glyceryl stearates
Glycol palmitate
Glycol stearate

Hexyl laurate
Isocetyl alcohol
Isodecyl stearate
Isohexyl palmitate
Isopropyl isostearate
Isopropyl myristate
Isopropyl palmitate
Isostearyl alcohol
Jojoba oil
Lanolin
Methyl palmitate
Myristyl propionate
Octyl Octanoate
Octyl stearate
PEG-4 lanolate
PEG-5 tristearyl citrate
Polyglyceryl-6 oleate
Polyglycerol-2 trisostearate
PPG-20 cetyl ether
PPG-4 laureth-2
Propylene glycol linoleate
Squalene
Sucrose oleate
Sunflower seed oil glycerides
Tall oil glycerides
Tridicyl stearate
Wheat germ glycerides

4. Miscellaneous

- These ingredients are sometimes added to moisturizers to create a special effect on the skin such as enhancing the appearance of dry or damaged skin.
- Chemicals that slow oxidation by reacting with free radicals include tocopherols and ascorbic acid.
- Citric acid, tartaric acid, and EDTA do not have strong antioxidating properties, but enhance the antioxidant effects of other ingredients.

Alkyl galate

Allantoin

Aloe barbadensis

Arachidonic acid

Ascorbic acid

Azulene

Beta-carotene

Betaglucan

Bioflavonoids

Biabolol

Butylated hydroxytoluene (BHT)

Butylene glycol

Caprooyl sphingosine

Capsaicin

Carnitine

Carnosine

Ceramides 1/2/3/4/5/1a/611

Cholesterol

Citric acid

Coco-betaine

Collagen

Dimethicone copolyol acetate

EDTA

Gelatin

Glucose

Glycosphingolipids

Hyaluronic acid

Hydrolyzed keratin

Hydrolyzed soy starch

Isodecyl salicylate

Lactic acid

Linoleic acid

Maltodextrin

Methyl nicotinate

Milk protein

Mineral salts

Niacin

Papain

PEF-5 tall oil sterol

Phospholipids

Retinol

Salicylic acid

Skin lipids

Superoxide dismutase

Tartaric acid

Theophylline

Tocopheryl acetate

Tyrosine

Ubiquinone

Undecylenyl alcohol

Zinc gluconate

Formulation Characteristics

- Most moisturizers are formulated as creams (water-in-oil emulsion) or lotions (oil-in-water emulsion).
- The "heaviness" of the final formulation correlates with the inclusion and relative concentration of heavier occlusive agents such as petrolatum and lanolin derivatives,
- It also depends on inherent qualities of individual emollients and humectants that may be included in some products, and the oil-water ratio.
- Night creams are examples of products that are specifically designed to be heavier formulations.
- Specific ingredients are often combined in formulations to correlate with use for individual "skin types" such as dry, normal, or oily complexions.
- This is achieved by altering the heaviness characteristic of the occlusive agent used through selection of specific emollients that may be either protective, fattening, dry, or astringent in their inherent quality, and through adjustment of oil-water ratios.
- Examples of ingredient adjustments designed to correlate with use in specific skin types include
- Dimethicone, a non-greasy, noncomedogenic emollient agent used in "oil free" facial moisturizers marketed for individuals with "oily skin"
- Inclusion of oil-absorbent compounds such as kaolin or talc, added to formulations to reduce "facial shine" by absorbing excess sebum.

Evaluation

1. Bioengineering measurement method
2. Static measurement
 - a. Short term test
 - b. long term test
3. Dynamic measurement test
4. Sorption desorption test
5. Moist accumulation test
6. Plastic occlusion stress test
7. Near IR based spectroscopic measurement

Using bioengineering measurement methods

Include measurements of electrical properties, spectroscopic methods such as IR absorption spectroscopy and emission, evaluation of the barrier function of the stratum corneum (SC), measurement of mechanical properties, transient thermal transfer, nuclear magnetic resonance imaging, skin surface topography, and scaling evaluation.

Most frequently, bioengineering techniques based only on the electrical properties of the SC together with measurement of transepidermal water loss (TEWL) are used.

Short-Term Tests/Single Application

- The tests are conducted most of the time on the inner side of the forearm of healthy subjects and allow a randomized side-to-side comparison of test products with a placebo or vehicle, a known active product, and an untreated control skin.
- Four to six products may be simultaneously tested. The products are applied at the rate of 2 mg/cm². Two different experimental designs may be used:

- This procedure better picks up the activity of a humectants contained in the test preparation, whereas the vehicle effect is strongly attenuated by the uniform conditions encountered under the occlusion patch.

Long-Term Tests/Multiple Applications

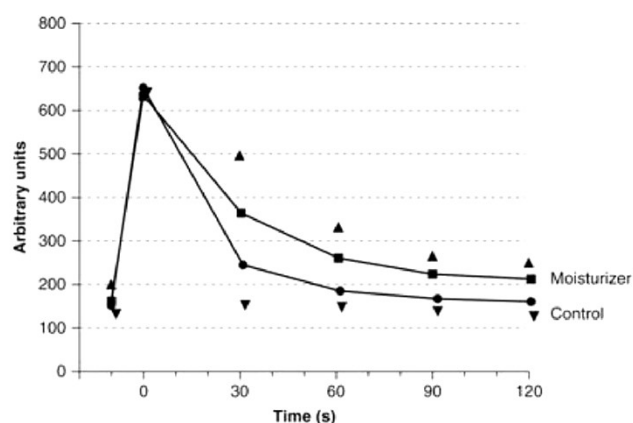
- The design of these tests and selection of subjects is similar to the regression method previously described but with a modified and shortened regression protocol.
- The treatment period extends over one week only, and the regression phase takes place over the following week.
- Bioengineering measurements are conducted 12 to 16 hours after the treatment or moisturizer application, and for the last time on the Monday following the regression week.
- Inclusion of these noninvasive measurements allowed rapid and reliable product performance evaluation.

Dynamic Measurements

- These tests, in addition to the classic evaluation of skin hydration, provide information on dynamic properties of the SC .
- These properties are likely to be modified by the humectants (e.g., glycerol, urea, α -hydroxy acids) incorporated in the moisturizers used for treatment.
- Generally speaking, dynamic function tests are characterized by the assessment of the skin's response to a given external stimulus that can be physical (e.g., water, occlusion, stretch, and heat) or chemical (e.g., drugs and irritants) in nature.
- These dynamic tests may be used either during short-term or long-term product testing, and will usually be performed before and at different time points after treatment.

The Sorption-Desorption Test

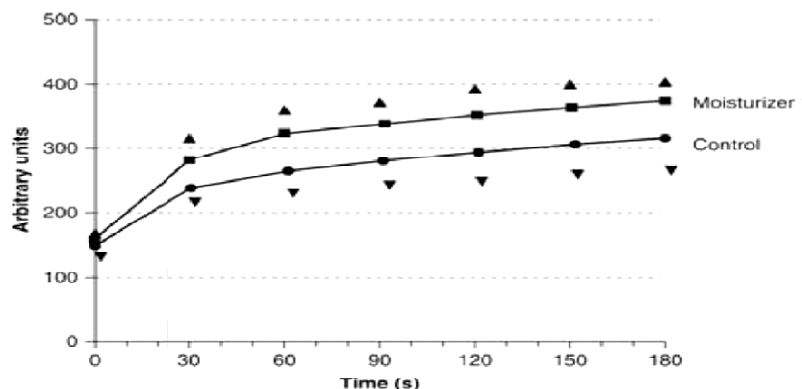
- This test gives information about the water-binding capacity of the uppermost layers of the SC
- It is best conducted using measurement devices that are able to measure hydration on a wet surface and that give instantaneous readings on contact with the skin.
- The first value represents the hydration state of the SC. Then 50 mL of distilled water is pipetted onto the skin, left in place for exactly 10 seconds, and wiped with a soft paper towel.
- Then hydration is immediately measured. Further measurements are taken at 0.5, 1, 1.5, and 2 minutes.
- Parameters such as hygroscopicity, water sorption capacity, water-holding capacity, and accumulated water decay may be calculated from the measurement curve and used to characterize the state of the SC and/or different properties of the tested products



Time course of hydration changes during a SDT performed 60 minutes after a single one-hour application of a moisturizer.

The Moisture Accumulation Test

- This test gives information about the quantity of moisture the SC may accumulate during a given time
- It is conducted with a device that can measure continuously after bringing the probe in contact with the skin surface.
- The probe then remains on the skin for three minutes, thereby creating occlusive conditions.
- The moisture accumulation test (MAT) measures the accumulation of water under the probe every 0.5 minutes. Water accumulation is evaluated by calculating the area under the time curve until three minutes.



Time course of hydration changes during a MAT performed 60 minutes after a single one-hour application of a moisturizer

Pigmentation

- Skin pigmentation has far reaching social and psychological implications.
- White people strive for tanning while brown and black people strive for a lighter skin.
- Pigmentation is mainly due to melanin.
- Melanin pigmentation disorders are important for medical and cosmetic reasons.

Normal skin pigmentation is influenced by:

1. The degree of vascularity
2. The amount and location of melanin
3. The presence of carotene
4. The thickness of the horny layer

Anti Pigmentation Creams

- Fabindia vitamin E D- pigmentation cream
- Shahnaz Husain Shalwhite pigmentation lotion
- Jovees Anti Blemish pigmentation cream
- Vedic Line Alpha Whitening De pigmentation cream
- Biotique Bio Morning Nectar lotion
- Kaya pigmentation reducing complex
- Avon Care Fairness cream

Prickly Heat

Prickly heat is also known as miliaria rubra is an itchy red spot or bump which develops when sweat glands under the skin get blocked because of which sweat can't escape and leaks out on skin resulting in small blisters or red bumps that you can see on your skin. It can also be caused by less than developed sweat glands as well as bacterial infections. From your groin, armpits and abdomen to your neck, back, hands and feet, prickly heat can appear anywhere on your body.

How to prevent prickly heat??

- Wear only clothes made of cotton. This will allow your skin to breathe, which in turn means lesser sweating.
- To avoid excessive sweat build up you must bathe regularly. However use gentle skin care products by a brand like Olay to avoid any skin irritation.
- Keep your body cool leave the part of the skin where you are experiencing prickly heat uncovered and exposed to cool air.

Prickly heat powder

Formula

- Boric Acid: 0.89gm
- Talc: 4.48gm
- Starch: 4.48gm
- Menthol: 0.178gm

Procedure

Weigh the required quantity of boric acid as well as talc mix with them well in a mortar. After mixing add starch and again mixing. Then add menthol and again mixing till fine powder made and no glittery particles.

Body Odor

When body gives off a scent others may find unpleasant it is known as body odor.

Body odor usually becomes evident if measures are not taken when a human reaches puberty. People who are obese those who regularly eat spicy foods as well as individuals with certain medical conditions such as diabetes are more susceptible to having body odor.

Body odor is most likely occur in the following spaces:

- Feet
- Groin
- Armpits
- Genitals
- Pubic hair and other hair
- Belly button
- Anus
- Behind the ears

Most commonly used body sprays for preventing body odor. Deodorants are also used.

Deodorant formulation - Solid stick form of deodorant that contains a polyol, glycerine at least one of potassium chloride and potassium sulphate, zinc hydroxide, magnesium hydroxide, sodium stearate, water, and a cosmetically acceptable filler. The magnesium hydroxide and zinc hydroxide are present in a molar ratio up to approximately 3:1

Wrinkle

- A wrinkle also known as a rhytide is a fold ridge or crease in the skin or on fabric. Skin wrinkles typically appear as a result of aging processes such as glycation, habitual sleeping position, loss of body mass or temporarily as the result of prolonged immersion in water.
- Anti wrinkle creams are most commonly used to prevent wrinkles.
- Anti wrinkle creams fight wrinkles and to reduces wrinkles and tone up facial skin.

Anti Wrinkle Creams

- Fresh lotus youth preserve moisturizer
- Oily regenerist micro sculpting cream
- Nourish cream
- Body merry retinol surge moisturizer
- Neutrogena

Common ingredients in anti wrinkle creams

1. **Retinol** - Retinol is a vitamin A compound the first antioxidant to be widely used in non prescription wrinkle creams.
2. **Vitamin C** - Vitamin C may help protect skin from sun damage. So wrinkle creams containing vitamin c must be stored in a way that protects them from air and sunlight.
3. **Hydroxy Acids** - Alpha hydroxyl acids beta hydroxyl acids and ploy hydroxy acids are exfoliants substances that remove the upper layer of old dead skin and stimulate the growth of smooth evenly pigmented new skin.
4. **Coenzyme Q10** - Which help to reduce fine wrinkle around the eyes and protect the skin from sun damage
5. **Peptide** - This used for in products for wound healing stretch marks and now wrinkles.

6. **Tea Extracts** - Green, black and oolong tea contain compound with antioxidant and anti inflammatory properties. Green tea extracts are most commonly used.
7. **Grape Seed Extract** - Grape seed promotes wound healing.
8. **Niacinamide** - It helps to reduce water loss in the skin and may improve skin elasticity.

Dandruff

Dandruff-medically termed pityriasis simplex/ capitis or furfuracea capitis is a chronic condition that is marked by skin flaking on scalp & itching. Term "dandruff" or pityriasis means bran-like epidermal scaling was introduced into the dermatological vocabulary by Galen. There are two types of dandruff that usually occur- common (dry/sicca) or oily/steatoides dandruff. The pathogenesis of dandruff is associated with *Malassezia furfur* (previously known as *Pityrosporum ovale*).

Dandruff has peak incidence and severity at about 20 years of age and becoming less frequent after the age of 50.

Causes of dandruff:

- Family history- strongly related to dandruff
- Neurological abnormalities including:
- Depression
- Parkinsonism

Lifestyle factors including:

- Incorrect skin care
- Micro trauma
- Stress
- Excessive perspiration

Diet factors including:

- Poor nutrition,
- Alcoholism,
- Food allergies

Environmental factors including:

- Pollution
- Exposure to dust
- UV light exposure
- Climate changes - Cold, dry winters cause dandruff or making it worse.

Hormonal factors include:

Increased sebaceous gland action (over activity) at the time of puberty. Young adults with increased androgen levels, which is triggered by stress

Immunological abnormalities

- HIV or AIDS with proportionality in 34-83% patients
- Increased sensitivity to M. furfur.

Cosmetic factors include:

- Excessive use of harsh shampoos
- Excessive use of hair dyes
- Use of alkaline soaps

Associated dermatological diseases:

Dry skin: during winter when the air is cold and rooms are overheated is the most common cause of dandruff

- Seborrheic dermatitis (Irritated, oily skin): This condition, a frequent cause of dandruff.
- Psoriasis: accumulation of dead skin cells resulting to silvery scaling.
- Contact dermatitis (sensitivity to hair care products)

Seborrhea dermatitis is also associated with:

- Pancreatitis
- HCV infection
- Neoplasms

Dandruff Treatment

1. Symptomatic Treatment

- Reduction of itching (pruritus)
- Reduction of scale formation (desquamation)
- It is often achieved by frequent hair washing with normal shampoo (without any active ingredients). The effect is due to the surfactants in the shampoo but is usually only transient.

2. Causal treatment

- Reduction of Malassezia
- Reduction of scalp inflammation
- Reduction of skin cells proliferation
- It includes the anti - squamolytic agents and anti-fungal agents repeatedly used.
- Anti-squamolytic agents include salicylic acid and coal tar which both reduce skin proliferation and promote desquamation.
- The most frequent antifungal agents are tea tree oil, piroctone olamine, zinc pyrithione, selenium disulphide, terbafine, and ketoconazole. All are active against yeast and reduce inflammation of the scalp

3. Prevention of Relapse

- Prevention of regrowth of *Malassezia*
- In order to avoid regrowth of yeast, we use anti-fungal shampoos at regular intervals (e.g. 1-2 times a month).
- Ketoconazole has demonstrated its superiority in case of antifungal action against *Malassezia*.
- Coal tar is most frequently being used
- Genetic abnormalities (Hailey-Hailey disease & Down syndrome)

Recommended formulas were classified into three groups on the basis of their mechanism of action.

1. Fungicidal substances (zinc pyrithionate, imidazole, ketoconazole)
2. Cytostatic substances (tar, selenium sulphide, octopirox)
3. Keratolytic substances (tar, salicylic acid, urea, sulphur compounds)
 - Cytostatic & keratolytic agents are for symptomatic management only & recurrence is observed after stopping the treatment. Therefore, these groups of agents are recommended in politherapy.
 - Octopirox is cytostatic agent with additional antibacterial and antimycotic activity.

Ketoconazole is an imidazole antimycotic agent and has been used orally for the treatment of multiple mycoses & efficacious against pityriasis capitis and seborrheic dermatitis. Ketoconazole binds and inhibits cytochrome P450 14-alpha-demethylase (enzyme required for the formation of ergosterol from lanosterol i.e. blocks fungal synthesis of ergosterol--- an essential constituent of fungal cell membranes).

Coal tar is approved for over-the-counter treatment of dandruff, seborrheic dermatitis and psoriasis reduces epidermal cells (size & number), epidermal proliferation and dermal infiltrates.

Salicylic acid is approved for over-the-counter treatment of dandruff, seborrheic dermatitis and psoriasis is an exfoliant that loosens weakly adherent flakes, enabling them to be washed away.

- Salicylic acid, liquor carbonis & sulfur have exfoliative qualities to improve the appearance of scaling, while the antimitotic effect of topical corticosteroids and coal tars might also be involved in reducing the hyperproliferation associated with dandruff scaling.

Zinc pyrithione (ZPT) is a biocide & it is only sparingly water-soluble, allowing efficient scalp retention after rinsing. This attribute has led to ZPT becoming the most common material used for dandruff treatment globally.

Tacrolimus has been shown to have potent antifungal activity against *Malassezia furfur* in vitro. Tacrolimus and pimecrolimus may be effective as they possess both anti-inflammatory and antifungal activity.

Anti dandruff shampoo

The shampoos simultaneously clean the hair and scalp by emulsifying oily secretions while treating the underlying disease. It is recommended to apply the shampoo, leave it on the scalp for 5-10 minutes, and then rinse. The shampoo is applied typically once a day fortnightly, then 1-2 times a week thereafter for maintenance.

1. Keratolytic Shampoos

The pathogenesis of dandruff involves hyperproliferation, resulting in deregulation of keratinization. Keratolytic agents (salicylic acid and sulphur) loosen the attachments between the corneocytes and which get washed away when shampooing.

a. Salicylic Acid

Salicylic acid is a beta-hydroxy acid keratolytic agent that decreases cell to cell adhesion between corneocytes & removes scaly, hyperkeratotic skin. It may involve the release of desmogleins and the disintegration of desmosomes & activation of an endogenous pathway for normal cell separation.

b. Sulfur

It is having both keratolytic and antimicrobial properties. The keratolytic action is mediated by the reaction between the sulfur and the cysteine in keratinocytes, & the antimicrobial effect occurs with conversion of sulfur to pentathionic acid by normal skin flora or keratinocytes.

2. Regulators Of Keratinization

a. Zinc

It is thought that zinc pyrithione (ZPT) acts by sebum production, normalizing epithelial keratinisation, or both. It also causes significant reduction in the numbers of yeast organisms.

Normalization of the stratum corneum ultrastructure by zinc pyrithione is secondary to the correction of the pathology in the living epidermal layers.

b. Tar

Although tars have been classically used to treat psoriasis but they are effective treatment option for dandruff also. Problems with odor, staining, and messiness in application makes tar a second-line therapy in most of the cases. Tar shampoos work through antiproliferative and cytostatic effects. These products disperse scales & reduce *Malassezia* colonization (suppresses epidermal DNA synthesis).

c. Steroids

Topical corticosteroids work via their anti-inflammatory and antiproliferative effects. On the scalp, lotions or solutions having moderate-to-high potency are typically used.

Clobetasol propionate 0.05% is available in a shampoo form. Topical steroids are often used in combination with other dandruff treatments such as antifungal agents.

3. Antimicrobial Agents

a. Selenium Sulfide - It controls dandruff via its anti-*Pityrosporum* effect & also reduces the cell turnover rate significantly. It produces a cytostatic effect & appears to have anti-seborrheic properties on cells of the follicular epithelium and epidermis.

A significant ADE is excessive scalp oiliness for many patients regularly using selenium sulfide for dandruff control.

b. Imidazole Antifungal Agents

Imidazole topical antifungals block the formation of ergosterol (essential component of the fungal cell membrane) resulting into decreased fungal growth and survival. Ketoconazole is an antimycotic agent (broad-spectrum) & active against both *M. furfur* and *Candida albicans*. Of all of the imidazoles currently available, ketoconazole has become a leading contender because of its effectiveness in treating seborrheic dermatitis among other treatment options. Rare ADEs include irritation and stinging. Various studies have demonstrated that ketoconazole was more effective than zinc pyrithione or selenium sulfide in reducing *Malassezia*.

c. Hydroxy pyridones

The hydroxypyridones interfere with the active transport of essential macromolecule precursors, cell membrane integrity, and cell respiratory processes of dermatophytes and yeasts. Ciclopirox 1% a member of the hydroxypyridone family, has broad-spectrum action against many fungal species. It also inhibits prostaglandin and leukotriene synthesis. Its anti inflammatory action is demonstrated in human polymorphonuclear cells.

Home Remedies for Dandruff

No doubt, natural remedies take time to show results but they are effective to treat the problem completely.

Home remedies for dandruff may include:

1. **Tea Tree Oil** - Very strong natural antiseptic used to treat dandruff. It can be mixed with the regular shampoo or may be put in a bottle with warm water and solution can be sprayed over.
2. **White Vinegar** - Vinegar has acetic acid that retards bacterial and fungal growth. One part vinegar mixed with three parts of water and then used it to wash your hair and rinsing with normal water or direct application on to hair(wrapping with a towel left for overnight) & washing hair with your regular shampoo can control dandruff to a great extent.
3. **Olive Oil** - It is a good moisturizer that keeps scalp away from drying and flaking. Slightly warm virgin olive oil is used to massage the scalp and then a towel is wrapped around head for half an hour. Finally thoroughly brushing the hair & washing hair properly helps get rid of the dry scales.
4. **Lemon** - Lemon due to its acids helps get rid of dandruff causing yeast.

Procedure includes taking fresh lemon juice and massaging scalp with it & then, leaving it for a few minutes and then wash hair with a shampoo followed by application of a conditioner. This remedy is repeated daily until dandruff vanishes. Else one tablespoon of lemon juice mixed with five tablespoon of coconut oil and then applied to hair and scalp with wrapping a towel around hair and leave it overnight and finally washing hair in the morning proves beneficial also.

5. Baking Soda

It controls dandruff causing overactive fungi. This starts with wetting hair and then rubbing some baking soda on scalp followed by rinsing hair properly after some time only with plain water (do not use shampoo).

With its repeated use, scalp starts producing natural oils & there are dry flakes left. Alternatively adding some baking soda to regular shampoo and then using it to wash hair is also practicable.

6. Fenugreek Seeds

Anti-fungal and scalp soothing properties of this herb help control dandruff to a great extent. The procedures includes soaking some fenugreek seeds in water overnight and in the morning grinding it into a fine paste & applying this paste all over the scalp with an interval of a few hours & finally washing hair properly with a mild shampoo. I this remedy is followed twice a week if there is severe dandruff problem.

7. Aloe Vera

It can control dandruff through its healing, antifungal, & antibacterial properties. Just a few minutes before taking a shower, some fresh aloe vera juice or aloe vera gel is applied to hair & then hair are washed with a mild shampoo. This remedy is repeated at least twice a day.

8. Thyme

Its anti fungal and antiseptic properties help dandruff treatment. Four tablespoons of dried thyme put in two cups of water and then boiled for 10-15 minutes & massaging scalp thoroughly with this solution after it cools down & then leaving it for five minutes & repeating the process again (hair are not washed for 2-3 hours).

9. Aspirin

Acetylsalicylic acid (the ingredient of aspirin) helps control dandruff.

Two aspirins crushed properly and mixed with equal amount of shampoo, then washing hair with this mixture and massaging the scalp with it. Then, leaving it for a few minutes and finally rinsing hair properly with plain water can help in controlling dandruff

10. Listerine

Its antidandruff effect might be due to the herbal oils and alcohol present in it. Using it to rinse hair & leaving it for 10-15 minutes and then properly washing hair can be helpful to get rid of dandruff.

Bleeding Gums

The main cause of bleeding gums is the buildup of plaque at the gum line. This will lead to a condition called gingivitis, or inflamed gums. Plaque that is not removed will harden into tartar. This will lead to increased bleeding and a more advanced form of gum and jawbone disease known as periodontitis.

Treating Bleeding Gums Naturally

1. Clove Oil - Clove oil helps to reduce the inflammation of gums and stops the gums from bleeding. Take a little clove oil and rub it on your gums or chew one or two cloves. You may feel a slight burning sensation, but it will reduce the inflammation to a greater extent.

2. Aloe vera - Aloe vera also reduces gum inflammation, making bleeding less likely. Take a small amount of aloe vera pulp and massage it on your gums. Let the pulp settle on your gums before rinsing your mouth. Mild gum diseases can be taken care of by consuming liquids that have natural aloe vera in it.

3. Fresh fruits and Vegetables

Fresh and raw vegetable and fruits are also equally beneficial for healthy gums. Fruits and vegetables are rich in vitamins, minerals and low in calories. Chewing raw vegetables help to improve blood circulation in gums, thus helping to reduce the bleeding of gums. Treating Bleeding Gums Naturally Salt Gargling One of the easiest and simplest home remedies is to gargle with salt water. Take some lukewarm water and add a bit of salt and gargle with it thrice a day. Treating Bleeding Gums

4. Consume Fruits rich in Vitamin C

Deficiency in vitamin C can be the cause of gum diseases like gingivitis etc. Fruit like orange, lemon, broccoli and cabbage are the best and rich sources of Vitamin C. Use plenty of them to make up for the deficiency. Alternatively, go for health supplement providing vitamin C.

5. Milk

Milk, which is rich in calcium can be used as an effective home remedy for bleeding gums. However, bear in mind that milk is also a source of fat. When you take milk, fat may stick to your gum which is not good for healthy gums. It is, therefore, very important to brush your teeth after taking milk. Dentists advise not to take milk one hour before going to bed. Treating Bleeding Gums Naturally

6. Baking Soda

Baking soda is another easy to use and readily available home remedy for bleeding gums. Dip your tooth brush in the baking soda powder and brush your teeth normally as you would do using toothpaste. Treating Bleeding Gums Naturally

7. Cranberry Juice

Cranberry Juice has anti-bacterial properties and can help in fighting against gum problems. Bear in mind that the cranberry Juice sold in the supermarkets is either too much diluted or mixed with sugar. It is best to have home made fresh juice. Treating Bleeding Gums Naturally

8. Black Seed Oil

It is also known as miracle cure to treat any condition. It has antimicrobial properties and fights against harmful bacteria. Brush your teeth and then massage your gums with a few drops of black seed oil. Treating Bleeding Gums Naturally

9. Garlic and Honey

Garlic is a very powerful anti-biotic. When combined with manuka honey, it forms a very powerful anti-bacterial tonic. Rub the garlic-honey paste on the gums. Please note that you will feel burning sensation. In order to reduce the burning sensation mix half a clove of garlic into one tablespoon of raw honey and apply the paste.

Mouth Odor

Halitosis is a general term used to define an unpleasant or offensive odour emanating from the breath regardless of whether the odour originates from oral or non-oral sources. Halitosis should not be confused with the generally temporary oral odour caused by intake of certain foods, tobacco, or medications

Treatment

- Self assessment tests(subjective tests)
- Whole mouth malodor (Cupped breath)

The subjects are instructed to smell the odor emanating from their entire mouth by cupping their hands over their mouth and breathing through the nose. The presence or absence of malodor can be evaluated by the patient himself/herself.

Wrist lick test - Subjects are asked to extend their tongue and lick their wrist in a perpendicular fashion. The presence of odor is judged by smelling the wrist after 5 seconds at a distance of about 3 cm.

Spoon test Plastic spoon is used to scrape and scoop material from the back region of the tongue. The odor is judged by smelling the spoon after 5 seconds at a distance of about 5 cm organoleptically.

Dental floss test - Unwaxed floss is passed through interproximal contacts.

Objective Tests

- Organoleptic measurement
- Gas chromatography (GC)
- Sulphide monitoring
- Electronic nose

- BANAtest
- Tongue costing index
- Dark Field or Phase Contrast Microscopy
- Saliva Incubation Test

Preventive Measures:

Preventive measures rather than curative aspects are highly recommended.

- Periodical tooth cleaning by dental professional.
- Brushing of teeth twice daily with appropriate brushing techniques and for duration of 2-3 mins.
- Use of a tongue scraper to get rid of the lurking odour causing bacteria in the tongue surface.
- Flossing after brushing to remove food particles stuck in between the tooth surfaces.
- Limit intake of strong odour species.
- Limit sugar and caffeine intake.
- Drink plenty of liquids.
- Chew sugar free gum for a minute when mouth feels dry.
- Eat fresh fibrous vegetables such as carrots

Sensitive Teeth through Sensitive Formulations

Tooth sensitivity is tooth pain due to a wearing away of the tooth's surface or gum tissue. The most common cause of sensitive teeth in adults is exposed tooth roots due to receding gums. Because these roots are not covered by enamel, thousands of tiny channels leading to the tooth's nerve center (pulp) are exposed. When heat, cold or pressure touches these channels, you feel pain.

Ignoring your sensitive teeth can lead to other oral health problems. This is especially true if the pain causes you to brush poorly, making you vulnerable to tooth decay and gum disease.

First and foremost, tell your dentist or hygienist.

- Sensitive teeth can usually be treated successfully. Your dentist may prescribe a brush-on fluoride gel or a fluoride rinse.
- You can also try low-abrasion toothpastes with formulations made especially for sensitive teeth.
- Ask your dentist which anti-sensitivity products are right for you.
- Be careful to brush properly or you can cause your teeth to wear away, making them sensitive.
- Overzealous brushing, the clasp of a partial denture, and braces can also lead to abrasion (loss of tooth surface).

Unit 5 - Herbal Cosmetics

What are Herbal cosmetics?

Herbal cosmetics are defined as the beauty products, which possess desirable physiological activities, such as skin healing, smoothening, appearance, enhancing and conditioning properties with the help of herbal ingredients.

These are the cosmetics which are prepared by using plant products having cosmetic actions.

Herbal cosmetics used for

a. Skin:-

1. Powders
2. Bath and cleansing products
3. Creams
4. Make-up preparations
5. Lotions
6. Deodorants

b. Hair:-

1. Shampoos
2. Beard softeners
3. Tonics
4. Shaving media
5. Hair dressing
6. Depilatories (hair removers)

c. Teeth and mouth: -

1. Tooth powders
2. Dentifrices
3. Mouth washes

Herbal Cosmetics for Skin

The herbal approach of proper skin care is principally based on three essential steps.

- Cleanse
- Nourish
- Moisturize

Whatever may be the type of skin, these three steps are performed to protect the skin from the constant effect of environment, stress and skin's natural process of cell degradation decay.

Herbal cosmetics for various types of skin

For dry skin

- Example of Herbs: Rubia Cardifolia (Manjista), Tulsi, Glycerrhiza glabra in sesame oil.
- Fruit face mask: Banana or avocado pulp.

For sensitive skin

- Example of Herbs: Curcuma longa, Azadiracta indica.
- Fruit face mask: Banana or pineapple pulp.

For oily skin

- Example of Herbs: Terminalia arjuna, Curcuma longa, mustaka in musterd oil, milk, sour milk,
- Fruit face mask: Strawberry or papaya pulp.

Herbal ingredients used in skin care

Latin name	Common name	Part used	Uses
<i>Acorus calamus</i>	Sweet flag	Rhizome	Aromatic, Dusting powders, Skin lotions
<i>Allium sativum</i>	Garlic	Bulb	Promotes skin healing, Antibacterial
<i>Aloe vera</i>	Aloe	Leaf	Moisturizer, Sun screen, Emollient
<i>Alpinia galanga</i>	Galanga	Rhizome	Aromatic, Dusting powders
<i>Avena sativa</i>	Oat	Fruit	Moisturizer, Skin tonic
<i>Azadirachta indica</i>	Neem	Leaf	Antiseptic, Reduce dark spots, Antibacterial
<i>Echinacea purpurea</i>	Echinacea	roots, stem, and leaves	Skin regeneration
<i>Centella asiatica</i>	Gotu kola	Plant	Wound healing, Reduce

Example

Cleansing Creams: A satisfactory cleansing cream is water in oil emulsion that melts at temperature of body and spreads readily over the skin.

Formula: chamomile cleansing creams

- chamomile flower 50 parts
- distilled water 500 parts
- lemon juice 5 parts
- sodium benzoate 2 parts

Types of herbal hair care products

1. Hair Oil:

The hair oils are used for dressings and nourishing the hair. This preparation is generally used to increase the growth of hair and to make them healthy.

Hair oil should have following properties:

- a. They should give luster (soft glow) to the hair.
- b. Retain them soft and flowing.
- c. Invigorate their growth.
- d. Keep the brain cool.
- e. Should not be sticky.

2. Hair Lotion: Hair lotion has a stimulating effect upon the hair follicles. They are generally perfumed with oil of rosemary as it possesses a good stimulating property.

3. Hair Colorants: These are preparations which are used for the coloring of the hairs. They enhance the attractiveness of gray hair. They are applied externally on the hair with help of a brush.

4. Shampoo: Shampoo is preparation of surfactant in suitable form liquid, solid or powder- when used under the condition specified will remove surface grease, dirt, and skin debris from the hair shaft and scalp.

Qualities of an ideal hair care product

It should:

- Protect the hair cuticle.
- Cleans without stripping natural oils.
- Replace lost protein, moisture, and nutrients. Condition without weighing down the hair.
- Even without porosity and prevent moisture loss. • Smooth abraded cuticle scales.
- Prevent intense drying from environment.

Herbal ingredients used in Hair care:

Latin name	Common name	Part used	Uses
<i>Aloe vera</i>	Aloe	Leaf	Moisturizer, Shampoos
<i>Bacopa monneri</i>	Brahmi	Entire herb	Hair growth, Good for sleep, Shampoos
<i>Arctium lappa</i>	Arctium lappa	Roots	Promotes hair growth.
<i>Cedrus dessential oilsdara</i>	Dessential oilsdar	Wood	Soaps, Shampoos
<i>Centella asiatica</i>	Gotu kola	Plant	Hair care, Darkening of hair, Hair oil
<i>Citrus aurantium</i>	Orange	Peel	Soaps, Shampoos
<i>Citrus limon</i>	Lemon	Peel	Prevents hair loss
<i>Eclipta alba</i>	Bhringraj	Plant	Promoting hair growth, Shampoos, Hair oil

Herbal ingredients in oral care

In many traditional cultures, there are no plastic-bristle brushes, rather, the use of herbal "chewing sticks" are common.

Chewing sticks are usually taken from plants, shrubs or trees with high anti-microbial activity. The ends of selected sticks are shredded and they are used to massage the gums and "floss" the teeth.

Young twigs of any of a number of Neem, oaks, willows have served for chewing sticks, and are still available today.

The commonly used herbal medications for oral and dental health can be classified based on their properties as:

- a. Plants with antimicrobial property** - Aloe, betel pepper (*Piper betel*), black pepper (*Piper nigrum*), clove (*Syzygium aromaticum*), coriander, (*Coriandrum sativum*), eucalyptus (*Eucalyptus globules*), garlic (*Allium sativum*), turmeric (*Curcuma longa*)
- b. Plants with antioxidant property** - Spinach, pepper, black tea, broccoli, green tea, carrot, potato tomato, blackberry, grape, olive, pineapple have potent antioxidant activity.
- c. Plants with analgesic property** - Capsicum (*Capsicum annum*), clove (*Eugenia caryophyllus*), ginger (*Zingiber officinale*), menthol (*Mentha piperata*), poppy (*Papaver somniferum*), tobacco (*Nicotiana tabacum*).

Some of the important herbs and their use in dentistry and oral care as follows:

1. Clove Oil (*Syzygium aromaticum*)

It has got excellent analgesic and antiseptic properties which inhibit growth of all disease causing bacteria

Uses: Relieve toothache, Treating bleeding gums.

2. Aloe Vera (*Geloe kumari*)

It is analgesic, antibacterial, antiviral, antifungal, and antiseptic in nature.

Used in: Toothpick injuries, chemical burns.

3. Tea Tree Oil (*Melaleuca alternifolia*)

- It has antiseptic and antifungal properties.
- Used in the treatment of Throat Irritation.

4. Garlic (*Allium sativum*)

- Its antibacterial activity depends on allicin produced by enzymatic activity of allinase
- **Uses:** Inhibits the growth of *Streptococcus mutans*, and therefore, can be used as an effective remedy in the prevention of dental caries.

5. Evening Primrose (*Oleum Oenothera biennis*)

- It has anti-allergic and anti-ulcer activity.
- Used In: Treatment of dental caries.

6. Neem (*Azadirachta indica*)

- **Used In:** preventing and healing gum diseases and other dental problems such as bleeding and plaque

7. Thyme (*Thymus vulgaris*)

- It is mainly composed of volatile oils namely phenol, thymol, and carvacrol
- **Used in:** it contains fluoride & used in toothpaste

8. Turmeric (*Curcuma longa* Linn.)

- It is an antiseptic, antibacterial, anti-inflammatory and painkiller.

Review of guidelines for herbal cosmetics by private bodies like cosmos.

COSMOS - introduction

COSMetics Organic and natural Standard

- Sets certification requirements for organic and natural cosmetics products in Europe.
- The standard is recognized globally by the cosmetics industry.

In 2002 five European organisations responsible for setting organic and natural cosmetics standards met at a trade show to share ideas for broader standards to be used globally. These five COSMOS members are:

- BDIH (Germany)
- Cosmebio (France)
- Ecocert Greenlife SAS (France)
- ICEA (Italy)
- Soil Association (Great Britain)

To stimulate processes for sustainable production and consumption, the organic and natural cosmetics sector is using some simple rules governed by the principles of prevention and safety at all levels of the chain from production of raw materials to the distribution of finished products.

These rules are:

- Promoting the use of products from organic agriculture, and respecting biodiversity.
- Using natural resources responsibly, and respecting the environment.
- Using processing and manufacturing that are clean and respectful of human health and the environment.

- Integrating and developing the concept of "Green Chemistry".

The documents published with the standard include:

- The Control Manual: describes how the control systems works
- The Labeling Rules: explains how to use COSMOS labels
- The Technical Guide: additional information

This Standard applies to cosmetic products marketed as organic or natural and cosmetic ingredients with organic content. To be certified these products must comply with the defined criteria on:

- Origin and processing of ingredients
- Composition of total product
- Storage, manufacturing and packaging
- Environmental management
- Labelling and communication
- Inspection, certification and control.

Origin and processing of ingredients

6.1- ingredients

- Water
- Minerals and ingredients of mineral origin
- Physically processed agro-ingredients
- Chemically processed agro-ingredients
- Other ingredients.

6.2 Calculation rules for organic percentage

Composition of total product

7.1 Rules for cosmetic products under organic certification

7.1.1 Ingredients - At least 95% of the physically processed agro-ingredients must be organic.

7.1.2 Total product

At least 20% of the total product must be organic

7.2 Rules for cosmetic products under natural certification

There is no requirement to use a minimum level of organic ingredients

7.3 Calculation rules for natural origin percentage

% natural origin of total = [weight of total product - weight of non-natural origin ingredients (appendix V.1) - weight of petrochemical moieties (appendix V.3) / weight of all ingredients] x 100.

Storage, manufacturing and packaging

8.1 Storage

8.2 Manufacturing,

8.3 Packaging

8.4 Fabrics

Accepted Materials	Non accepted materials
Wood	materials Polyvinyl chloride (PVC) and other chlorinated plastic
Aluminium	Polystyrene
PE, PET, PP, PETG	Materials or substances that contain, have been derived from, or manufactured using, genetically modified organisms
Glass	
Paperboard	
PLA (non GMO) Any other 100% natural origin	

Environmental management

9.1 Environmental management plan

9.2 Cleaning and hygiene

Following disinfection materials can be used:

1. Iso-propyl alcohol
2. amphoteric surfactants
3. hydrogen peroxide
4. mineral acids and alkalis
5. peracetic acid (and stabilising agents)
6. formic acid
7. ozone

Labelling & communication

10.1 General rules - should be clear and non misleading to the consumer.

10.2 For products under organic certification must be labelled with the signature 'COSMOS ORGANIC' in conjunction with the seal of the COSMOS-standard AISBL member organization.

The product must not be called "organic", for example, "organic shampoo", unless it is at least 95% organic, measured as a percentage of the total product.

10.3 For products under natural certification must be labelled with the signature 'COSMOS NATURAL' in conjunction with the seal of the COSMOS-standard AISBL member organization.

10.4 For ingredients with organic content may be labelled with the signature 'COSMOS CERTIFIED'.

10.5 Supporting literature.

10.6 Organic in the name of a company or product range.

10.7 Use of the signature, name or term related to this Standard.

Inspection, certification and control

11.1 Inspection and certification

To be certified for cosmetic ingredients or cosmetic products under natural or organic certification according to this Standard, it is required to:

1. Have subjected the manufacturing operation and the ingredients or products to inspection and certification by an authorized certification body; this includes any sub- contracted manufacturing plants that are used to process the ingredients or products
2. Hold a valid operational certificate from that body
3. Undergo an on-site annual inspection cycle which may include possible unannounced inspections and other additional inspections, and
4. Allow the certification body to take samples and carry out analysis using laboratories conforming to ISO/IEC 17025 for ingredients or contaminants either on a random basis or in case of suspicion.

11.2 Approval of ingredients

To be certified for cosmetic ingredients or cosmetic products under natural or organic certification according to this Standard, it is required to:

- have subjected the manufacturing operation and the ingredients or products to inspection and certification by an authorized certification body; this includes any sub-contracted manufacturing plants that are used to process the ingredients or products hold a valid operational certificate from that body.

- Undergo an on-site annual inspection cycle which may include possible unannounced inspections and other additional inspections, and
- allow the certification body to take samples and carry out analysis using laboratories conforming to ISO/IEC 17025 for ingredients or contaminants either on a random basis or in case of suspicion.

11.3 Certification bodies Bodies certifying to this Standard must:

- be (associate) members of the COSMOS-standard AISBL
- be accredited to ISO/IEC Guide 65 (for any scope)
- be authorized according to the requirements set out in the COSMOS- standard Control Manual
- submit to annual authorization according to the COSMOS-standard Control Manual including any investigations that may be undertaken in case of complaint or suspicion, and
- Cooperate with other authorized certification bodies to ensure common interpretation and implementation of this Standard. Certification bodies must use the COSMOS-standard, and the COSMOS signatures, names and terms only in accordance with the requirements set out in this Standard, the Control Manual and the Labelling Guide, or otherwise only with the prior written consent of the COSMOS-standard AISBL.

12. Implementation

Guidelines with respect to preservatives:

- Cosmetic preservatives allowed by Ecocert and COSMOS are considered more skin friendly than some traditional preservatives.

- Most are naturally occurring in botanicals and so are considered by the certifying bodies to be suitable for products marketed as natural.
- According to the latest Ecocert and COSMOS standards the following preservatives are allowed in cosmetic products certified as Natural and Organic:

Ecocert / COSMOS Compliant Cosmetic Preservatives

Preservative	Ecocert / COSMOS compliant
Benzoic acid and its salts	Yes
Benzyl alcohol	Yes
Dehydroacetic acid and its salts	Yes
Salicylic acid and its salts	Yes
Sorbic acid and its salts	Yes

Regulations for Emollient

COMMERCIAL NAME / NON COMMERCIAL	EMOLLIENT EXAMPLES
1,3-Butylene glycol	Decyl glucoside
Acai oil	Eutrepe oleracea fruit oil
Acide stearique tp18/55	Palmitic and stearic acid
Acitire	Jojoba esters
Activonol -3	propanediol
akogel	Hydrogenated vegetable oil
Adeps lanae type sp PHEur	Lanolin
Aecosil	Glyceryl olivate Estolides

Guidelines for foaming agent:

Commercial name	Chemical name
PALMEROL 1299 - Lauryl Alcohol - RSPO SCC Mass Balance (MB)	Lauryl Alcohol
PALMEROL 1299 - Lauryl Alcohol	Lauryl Alcohol
PALMEROL 1498 - Myristyl Alcohol - RSPO SCC Mass Balance (MB)	Myristyl Alcohol
PALMEROL 1498*	Myristyl Alcohol

Regulations for emulsifying agents

COMMERICAL NAME/ NON COMMERICAL	EMULSIFIER
Akoline PGPR	Cetearyl alochol
ALBIWAX	Prunus armeniaca
Cholestrol BP	Chloestrol
Cosphaderm GMUD	Glyceryl undecylenate
CremerGLYC Refined glycerine.99.75%	Glycerin
Axol C62 pellet	Glyceryl stearate citrate
Dermofeel NC	Glyceryl stearate citrate0
Cholestrol EP	Cholestrol

Challenges

A key challenge is to objectively assess conflicting toxicological, epidemiological and other data and the verification of herbal materials used.

Unlike conventional cosmetics, there is extremely insufficient documentation of herbal cosmetics. Most herbal cosmetics depend on traditional methods or Ayurvedic formulations in their manufacturing. Some of these are passed down through generations, while some are documented. Even when documented, few of these can compare to the extensive clinical, pharmacological, and toxicological documentation available for conventional beauty products.

Herbal ingredients need special care and precautions during the complete phase of extraction process. Proper drying conditions should be adhered. Improper drying may lead to unintentional adulteration.

E.g. if digitalis leaves are dried above 65 °C decomposition of glycosides by enzymatic hydrolysis.

No specific standardization: there is no single standard that will apply to herbal cosmetic formulations, so no any good way to decide whether ingredient can be used or not, to which extent and what standards.

- Since the processing of one batch often differed from another, a single company's own product line may not have a standard output.
- This makes compliance with certain rules difficult, especially in some developed markets.

Few ingredients: Only few ingredients of natural origin can be used as preparation is nature based cosmetics. No synthetic surfactant, few preservatives can only be used. So there is uncertainty in best product preparation.

The problem of formulating with fewer ingredients may lead to the formation of product that may not work as well with synthetic chemical based competition. Consumers shall do some compromises for the fact that they are using natural based products but do not want to give up functionality for the sake of naturalness or sustainability as this is the beauty business.

Stability problem: One of the main problems faced in formulating herbal cosmetics preparation is the stability of herbal component both prior and after formulation as they are to be degrade easily by many physiochemical process.

- Deterioration happens especially during storage, leading to the loss of the active ingredients, production of metabolites with no activity and, in extreme cases, the production of toxic metabolites.
- Physical factors such as air (oxygen), humidity, light, and temperature can bring about deterioration directly or indirectly.

Safety and efficacy assessment has to be developed on each batch of manufacturing cycle thus the task become more tedious and time consuming.

From the manufacturer points of view, raw materials in bulk quantity and quality are hard to obtain which leads breakage in continuous production cycle of herbal cosmetics.

- Raw materials obtained from different place with varying climate cannot be treated as same as there arises difference in herbal potency.
- No specific criteria are given which may lead to irregularity in production cycle.

Poor quality of raw materials: Herbal cosmetics require the careful cultivation of plants and herbs in their natural environment.

- The raw material for herbal cosmetics company cannot be mass produced in a factory. Neither can one reproduce these in a matter of minutes in a laboratory.

Constrains of Herbal formulation

A. Constrains associated with handling of medicinal plants:

- Indiscriminate harvesting and poor post-harvest treatment practices.
- Lack of research on development of high-yielding varieties, domestication etc
- Poor agriculture and propagation methods.
- Inefficient processing techniques leading to low yield and poor quality products
- Poor quality and control procedures
- Lack of cGMP for preparation
- Lack of R&D on product and process development.
- Difficulties in marketing.
- Lack of trained personnel and equipment.
- Lack of facilities to fabricate equipment locally.
- Lack of access to latest technological and market information

B. Constraints associated with the dealing of Herbal Medicines:

Both the raw herb and the extract contain complicated mixtures of organic chemicals which may include fatty acids, sterols, alkaloids, flavonoids, glycosides, saponins, tannins and terpenes as well as other small molecules such as peptides and oligosaccharides.

- It is often difficult to determine which component, if any, of the herb has biological activity in humans.

Adulteration of herbal products can be made in various ways; commonly, adulteration is made by substituting other easily available or cheap plant species or sometimes by spiking of a product with synthetic constituents.

In addition, the processing of herbs, such as heating or boiling, may alter the dissolution rate, or even the functional activity of the organic constituents. Similarly, a host of environmental factors, including soil, altitude, seasonal variation in temperature, atmospheric humidity, length of daylight, rainfall pattern, shade, dew, and frost conditions, may affect the levels of components in any given batch of an herb.

Other factors, including insects, planting density, competition with other plant species, seeding time, and genetic factors, play an important role.

Plant collection for the use in botanicals is one of the factors of concern for quality. Plants collected in the wild may include non-targeted species, especially either by accidental substitution or intentional adulteration.